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Evaluation of a novel fish cell line co-culture system through PFAS-induced transcriptional responses

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Abstract

Ecotoxicology is increasingly challenged by the demand for toxicological data on a growing range of environmental contaminants. At the same time, continued reliance on laboratory animals is limited by ethical considerations and regulatory principles. The development of novel cell-based methodologies, replacing and reducing the need for animal-based assays are therefore of vital importance for the continued progress and efficacy of ecotoxicological research and regulatory work.

In this study a co-culture model combining the two rainbow trout cell lines RTgutGC and RTL-W1 has been established in a dual compartment system. The aim was to evaluate this novel co-culture model's utility for future ecotoxicological applications. Following the establishment the model was exposed to perfluoroalkyl acids, most notably perfluorooctanoic acid (PFOA), and the transcriptional responses were recorded by real time quantitative PCR (qPCR). Additionally, a feeding regime in which rainbow trout was feed PFOA-spiked food was developed to enable comparisons to *in vivo* data. The study revealed both promising features as well as limitations for the *in vitro* model

PFOA exposure induced significant transcriptional responses related to lipid metabolism and xenobiotic detoxification processes, both in the co-culture model and in monocultures of RTL-W1. However, more pronounced treatment related effects were found for the monoculture, suggesting a lower sensitivity in the co-cultured cells. The absence of transcriptional responses for RTL-W1 in the basolateral compartment indicated low toxicant exposure, likely explained by low permeability of the gut cell barrier above. Comparisons to the dietary exposure data was proven challenging, highlighting the complexity of *in vitro* to *in vivo* extrapolations.

Interestingly, an elevated basal expression of CYP1A1 in the co-cultured RTL-W1 cells imply enhanced xenobiotic metabolic competence potentially triggered by the dual cell culturing environment. This finding shows potential for the model's applicability as an improved *in vitro* model for mechanistic studies of toxicological processes related to biotransformation reactions.

Sammanfattning

Ekotoxikologin står inför en växande utmaning i att tillgodose behovet av toxikologiska data för ett ökande antal miljöförorenande ämnen. Samtidigt begränsas användningen av djurförsök av etiska och regulatoriska skäl. Utvecklingen av nya cellbaserade metoder för ekotoxikologiska studier är därför avgörande för den fortsatta utvecklingen av forskning och riskutvärderande regulatoriskt arbete.

I den här studien etablerades ett dubbelkultur-system för två cellinjer, RTgutGC och RTL-W1 från regnbågsforell. Systemet bestod av två kammare skapade genom att cellkulturinsatser placerades i brunnarna på cellkulturplattor. Målet var att utvärdera möjligheterna för framtida användning av cellkultursystemet i ekotoxikologiska studier.

Efter att systemet etablerats exponerades det för perfluoroalkylsyror, primärt perfluorooktansyra (PFOA) och den transkriptionella responsen mättes med hjälp av kvantitativ PCR (qPCR). Dessutom utvecklades en metod för att exponera regnbågsforeller via PFOA-haltig mat. Detta för att möjliggöra jämförelser med de cellbaserade modellerna. Studien påvisade både möjligheter och begränsningar för den nya *in vitro* modellen.

Signifikant respons uppmättes för både dubbelkulturer och enkelkulturer av RTL-W1. Däremot var effekterna generellt mer tydliga i enkelkulturerna vilket talar för lägre känslighet för cellerna i dubbelkulturerna. Avsaknad av transkriptionella effekter för RTL-W1 i dubbelkulturernas nedre kammare tyder på låg exponering, sannolikt på grund av låg genomsläpplighet hos RTgutGC tarmbarriären ovanför. Jämförelser med *in vivo* data visade sig vara utmanande vilket påvisade svårigheterna med direkt extrapolering mellan cell- och djurmodeller.

Ett förhöjt basalt uttryck av CYP1A1 för RTL-W1 från dubbelkulturer jämfört med enkelkulturer indikerar ökad aktivitet i system för xenobiotisk metabolism. Den ökade aktiviteten kan ha utlösts av det dubbla cellkultursystemet. Detta visar att det finns potential för användning av systemet som en förbättrad *in vitro*modell för studier av cellaktiviteter kopplade till biotransformationsreaktioner.

Popular science summary

A new generation of laboratory methods based on fish cells instead of live fish could make it easier and more ethical to evaluate the risks of environmental contaminants.

As the list of chemical contaminants identified in the environment grows, so does the concern for their harmful impacts. We therefore need to increase our understanding of these chemical´s toxic effects and their spread in nature. However, increasing demand for more data clashes with efforts to reduce the chemical testing on animals. Development of new and improved laboratory methods that do not require continuous sacrifice of animals will therefore be vital if we want to protect the environment long term.

In this thesis project a promising new laboratory method based on fish cells is evaluated for future use in testing of environmental contaminants. Both limitations and future potentials for the new test setup were identified. In the system two different, so-called cell lines are grown together. These cell lines stem from real fish tissue but have been developed to enable continuous cell growth. The new cell system was tested with compounds from the infamous PFAS class, known for their persistence and health risks.

The cell responses in the new setup were measured on gene expression level and was then compared to PFAS responses in simple, more traditional cell-based methods. Additionally, live fish were feed PFAS spiked food and the gene expression in the fresh tissue was compared to the cell line responses. Direct comparisons were, however, proven to be challenging.

Heightened expression of some genes was found for the cell lines grown in the new setup which suggests increased activity in cellular systems that help break down toxic substances. These elevated activity levels are similar to what was measured in live fish. This shows potential for the new cell line system to be used as an improved non-animal-based test system suited for studies of toxicity responses in fish.

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List of abbreviations

RT – Rainbow Trout

FBS – Fetal bovine serum

PFAS - Per- and polyfluoroalkyl substances

PFAA - Perfluoroalkyl acid

PFECA – Perfluoropolyether carboxylic acid

PFCA - Perfluoroalkyl carboxylic acid

PFSA - Perfluorosulfonic acid

NAM - New approach methodology

TEER - Transepithelial electrical resistance

EC50 - Effect concentration 50%

Background

1. Ecotoxicology and the use of laboratory animals

Ecotoxicology, or ecological toxicology, is a scientific discipline studying the effects, interactions and fate of environmental chemical pollutants. Ecotoxicological research span all levels of biological organisation, from biochemical processes within cells to population level changes and ecosystem wide effects. Test subjects could be species collected in the field or standard laboratory pet species, commonly algae, crustaceans and fish. In addition to exploratory field- and laboratory studies ecotoxicology is closely linked to chemical policy making and safety testing of chemical products. Since the entry of the European union´s (EU) chemical regulation REACH (Registration, Evaluation, Authorization and Restriction of Chemicals) (EC 1907/2006), the risk assessment requirements for producers and importers of chemical products in Europe has gradually increased. Risk assessment of chemicals calls for standardised toxicological data and often requires the sacrifice of laboratory animals.

1.2 The ethics of animal studies

The continued utilization of laboratory animals for regulatory and scientific purposes is controversial and an ethical quandary for the scientific community (AKHTAR, 2015; Langan et al., 2024). The guiding principle for animal testing in the EU is to minimise the use of animals to solely necessary tests where no alternative methods are available, as stated in the directive on the protection of animals used for scientific purposes (Directive (EU) 2010/63). The basis for this approach was described in 1959 by Russell and Burch in their book *Principles of Humane Experimental Technique* where they present the principle of the three Rs (replace, reduce and refine). This principle advocates for the replacement of animal-based tests, reduction in the number of animals used, and refinement of animal welfare.

1.3 Laboratory animal use in the EU

Despite the aim to minimise use 8.5 million animals were utilised for research purposes in the EU and Norway in 2022, according to a summary report from the European commission (EU commission, 2024). 13% of these animals were used for regulatory purposes and the rest for basic research, applied studies and routine uses like antibody extraction and production of blood-based products. Overall, rodents are the most commonly used laboratory animals, followed by fish. The 2022 report describes a generally decreasing trend in numbers of animals used for scientific purposes with the main improvement stemming from methodological improvements in quality testing of medical products. Contrasting the overall trend, however, an increase in the use of Zebra fish (*Danio rerio*) was noted (EU commission, 2024).

2. New approach methodologies

Conducting *in vitro* studies is one approach to replacing and reducing the testing on laboratory animals. Studies conducted *in vitro* ("in glass") utilises isolated biological units outside of its biological context. It could be cells, microorganisms or biomolecules studied in, for example, a culture dish. *In vitro* studies are included in the abbreviated term NAMs, standing for new approach methodologies or non-animal methods.

Computational models like quantitative structure-activity relationships (QSARs) are other types of NAMs that already play a role in reducing the need for animal testing for risk assessments (ECHA, 2016; NAFTA, 2012). The development of reliable computational models requires large quantities of sound experimental data. High throughput *in vitro* methodologies could provide such data and assist in the development and improvement of *in silico* approaches. Interplay between NAMs could thereby help pave the way for biological research less reliant on laboratory animals.

In addition to serving as alternatives to animal-based studies NAMs are vital for providing toxicity data for the ever-growing list of chemicals manufactured and used globally. Wang et al. estimated in a 2020 publication 350,000 chemicals and chemical mixtures to be registered on the global market. Despite their registration the understanding of the toxicological risks of a majority of these substances, their intermediates, degradation products, and mixtures is insufficient (Kristiansson et al., 2021; Z. Wang et

al., 2020). Ensuring improved data quality and quantity, encompassing the whole chemical spectrum is vital to enable proactive policy making that protects both human health and the environment (Kristiansson et al., 2021). Relying solely on animal-based studies to increase the quantity of publicly available toxicological data is both unfeasible and in conflict with the principle of the three Rs. The development and improvement of NAMs are therefore of importance for the continued progress and efficacy of ecotoxicological research and regulatory work.

2.1 *In vitro* approaches in ecotoxicology

2.1.1 Continuous fish cell lines

Fish cell lines, also referred to as piscine cell lines from now on, are the basis of multiple established *in vitro* methods utilised in ecotoxicology. In contrast to primary cultures, which are short lived cultures from fresh tissue, cell lines can in theory be cultured continuously for an unlimited time (Mukherjee, 2023). Cell lines exist for a multitude of fish species and from a diverse set of tissue types. Cell lines originating from Rainbow trout (*Oncorhynchus mykiss*) is currently the most extensive piscine collection, with 79 listed cell lines in the Cellosaurus data base (https://www.cellosaurus.org/) as of March 2025 (Bairoch, 2018). Beyond their relevance in aquatic toxicology fish cell lines are also advantageous to mammalian cell lines as they are adaptable to a wide temperature range, tolerant to hypoxic conditions and have an overall slower metabolic rate. This allows for less frequent care and sustained culturing (Nguyen et al., 2024a).

Cell lines originate from primary cultures and are derived from cells that either through spontaneous mutations or via intentional processes acquire mutations that disrupt normal cellular senescence (Kaur & Dufour, 2012; Mukherjee, 2023). Compared to a primary cell culture the maintenance and use of an established cell line is simple and does not require the continued sacrifice of animals for fresh tissue samples. Cell lines enable high throughput test systems with unlimited, genetically identical material and produces highly reproducible results, comparable both within and between labs. Additionally, the number of repeats, replicates and treatment groups can be scaled up to ensure robust statistical results on a wide range of test parameters, without ethical constraints (Kaur & Dufour, 2012).

Cell lines have successfully been utilised in toxicology for deciphering relative toxicity of chemicals, which is helpful for focusing research efforts into substances of concern (Tanneberger et al., 2013). However, a monolayer of cells grown on polystyrene plastic lack its native environment and structure present *in vivo*. Moreover, continuous cell lines lose their tissue specific phenotypes, metabolic characteristics, their polarity and ability to form tight junction barriers (de Dios-Figueroa et al., 2021; Holgersson et al., 2024; Nguyen et al., 2024b). The physiological relevance of cellular responses from these de-differentiated cells could consequently be questioned (Kaur & Dufour, 2012). Extrapolating absolute toxicity and xenobiotic responses to whole organisms is therefore an ongoing challenge (Hernández-Moreno et al., 2022; Nguyen et al., 2024a; Tanneberger et al., 2013).

2.1.2 Improved and specialised cell culturing

Strategies to overcome some of the limiting factors of two-dimensional (2D) cell culturing is to create a more structurally complex culture, thereby enabling a stimulating *in vivo*-like microenvironment. Cell lines have successfully been cultured in three-dimensional configurations in two general ways, either as self-assembling spheroids or organoids, or in scaffolds designed to mimic the extracellular environment (de Dios-Figueroa et al., 2021; Nguyen et al., 2024a). Piscine cell lines from a few different species have successfully been grown as spheroids (Gómez-Mercader et al., 2024). *O. Mykiss* cell-based spheroids have been shown to exhibit improved xenobiotic metabolic competence compared to 2D cultures, and some recovered differentiated ultrastructural features have been observed (Hultman et al., 2019; Lammel et al., 2019; Uchea et al., 2015).

Dual compartment co-culturing of cell lines is another enhanced culturing system that could offer a more stimulating microenvironment. Co-culturing is the practise of culturing two or more cell populations in a shared environment, allowing for cell-cell interactions. It is a hybrid approach aiming to combine the ease of 2D culturing with the benefits of an enriched cellular environment. Chemical signalling and

exposure to excretion products in the medium shared between the cell populations has been proposed to induce de-differentiation of the cells and prompt more *in vivo*-like responses (Goers et al., 2014).

Additionally, dual compartment co-cultures allow for the creation of artificial systems that mimic physiological tissue and exposure pathways. While co-culturing of human cell lines has been used for multiple applications in biotechnology and pharmacological research similar opportunities for environmentally relevant species has been relatively underutilized (Goers et al., 2014; Kaur & Dufour, 2012; Langan et al., 2024). Co-culturing of fish cell lines is a promising technique for studying cellular pathways, tissue specific xenobiotic metabolism and intercellular interactions in response to environmental contaminants.

2.1.3 Gastrointestinal barrier models

Toxicant uptake via the gastro-intestinal barrier is one of the major exposure routes for fish, together with direct exposure across the gills. The intestinal barrier is a key system for toxicokinetic processes and vital for understanding system wide effects of toxicants (Minghetti et al., 2017; Vancamelbeke & Vermeire, 2017). An *in vitro* model of the human intestinal epithelium was developed as early as 1989 with the Caco-2 cell line grown on porous supports (Hidalgo et al., 1989). Studies of the intestinal barrier of environmental species has, however, traditionally been limited to *in vivo* or *ex vivo* studies. One established *ex vivo* method is the everted gut sack model, a method based on animal intestines incubated everted on a glass rod (Alam et al., 2012).

In the last decade promising intestinal epithelial models featuring the rainbow trout gut cell line RTgutGC, has been presented. In these models the cell line is grown on porous supports in a dual compartment system. The two compartments mimic the different nutritional environments of the intestinal lumen and the nutrient rich connective tissue lining on the opposing side of the epithelium (Vancamelbeke & Vermeire, 2017). This setup allows for polarisation with distinct upper (apical) and lower (basolateral) compartments and tight junction formation (Geppert et al., 2016; Minghetti et al., 2017; J. Wang et al., 2019). The model setup is pictured below in figure 1.

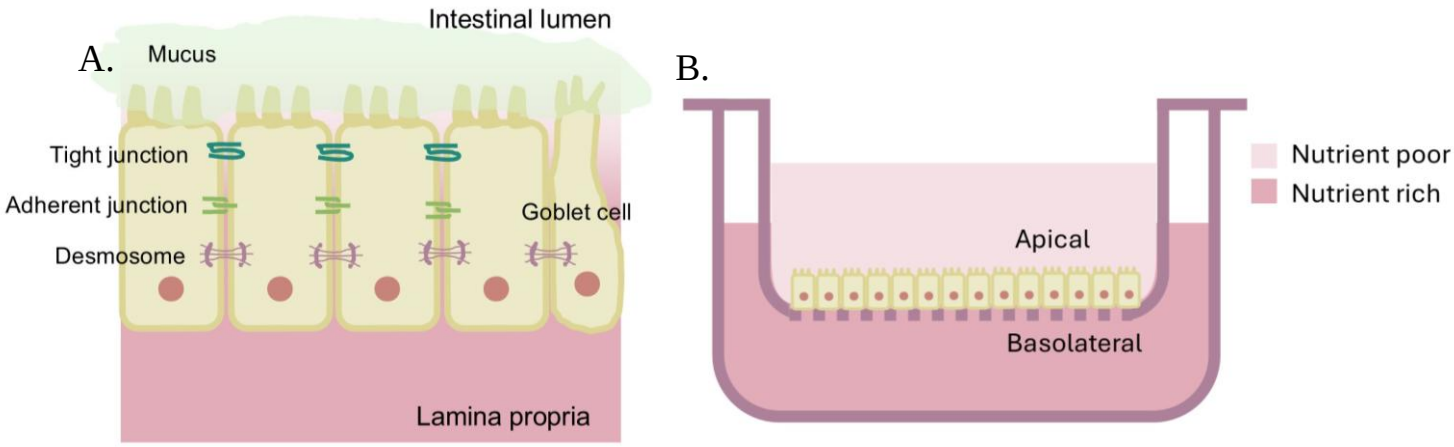

Figure 1, Schematic cross section of the intestinal epithelium and the barrier model on transwell insert. Figure created in Microsoft Powerpoint. **A.** Simplified schematic of the intestinal epithelium. Epithelium cells make up the main part of the barrier between the intestinal lumen and the connective tissue lining Lamina propria. Three kinds of cell-cell junctions hold the cells together: tight junctions, adherent junctions and desmosomes. Goblet cells produce mucins that line the apical side of the membrane. Figure based on Vancamelbeke & Vermeire (2017). **B.** Gut cells grown on an insert placed in the well of a cell culturing plate. The porous material at the bottom of the inserts allows for the cells to be in contact with two discrete medium compartments.

Combining this gastrointestinal barrier model with an additional cell population in the basolateral compartment creates a novel co-culture system. The RTgutGC barrier could for example be cultured together with cells from the rainbow trout liver cell line (RTL-W1) to mimics the interface between the intestinal compartment and the liver. This would allow for toxicokinetic studies of the gastrointestinal - liver exposure route. Additionally, more *in vivo*-like responses induced by cell signalling between the

two cell populations may improve the systems suitability for mechanistic studies. Ultimately this co-culture setup could help bridge the gap between simple studies based on monocultured cell lines and more demanding *in vivo* experiments.

3. Forever chemicals

Per- and polyfluoroalkyl substances (PFAS) encompasses a wide and diverse range of mainly man-made fluorinated hydrocarbons. Due to their amphiphilic structure with a hydrophilic head group and a hydrophobic fluorinated carbon chain they are both oil and water repellent. Additionally, the strong carbon-fluor bonds make the molecules highly stable and resistant to both biological and chemical degradation (C. Liu et al., 2007; Sunderland et al., 2019; Wee & Aris, 2023). These properties have made PFAS useful in a multitude of product like flame retardants, coatings, surfactants, lubricants and pesticides among many other (Buck et al., 2011; Glüge et al., 2020; Wee & Aris, 2023). Unfortunately, these chemical characteristics also make PFAS highly persistent in biological systems and in the environment, prompting the ominous nickname “forever chemicals” (Ahrens & Bundschuh, 2014; Wee & Aris, 2023).

3.1 Classes of PFAAs

Perfluoroalkyl acids (PFAAs) are divided into subgroups based on their chemical structure. Per-fluorinated compounds have a fully fluorinated carbon backbone while polyfluorinated compounds are partly fluorinated. The two main groups of non-polymeric per-fluorinated PFAAs are perfluoroalkyl carboxylic acids (PFCAs), characterized by a carboxylic head group, and perfluorosulfonic acids (PFSAs) with a sulfonic acid head group (Buck et al., 2011). The previously mentioned subgroups are presented in figure 2 with PFOA and PFOS as examples of PFCAs and PFSAs respectively.

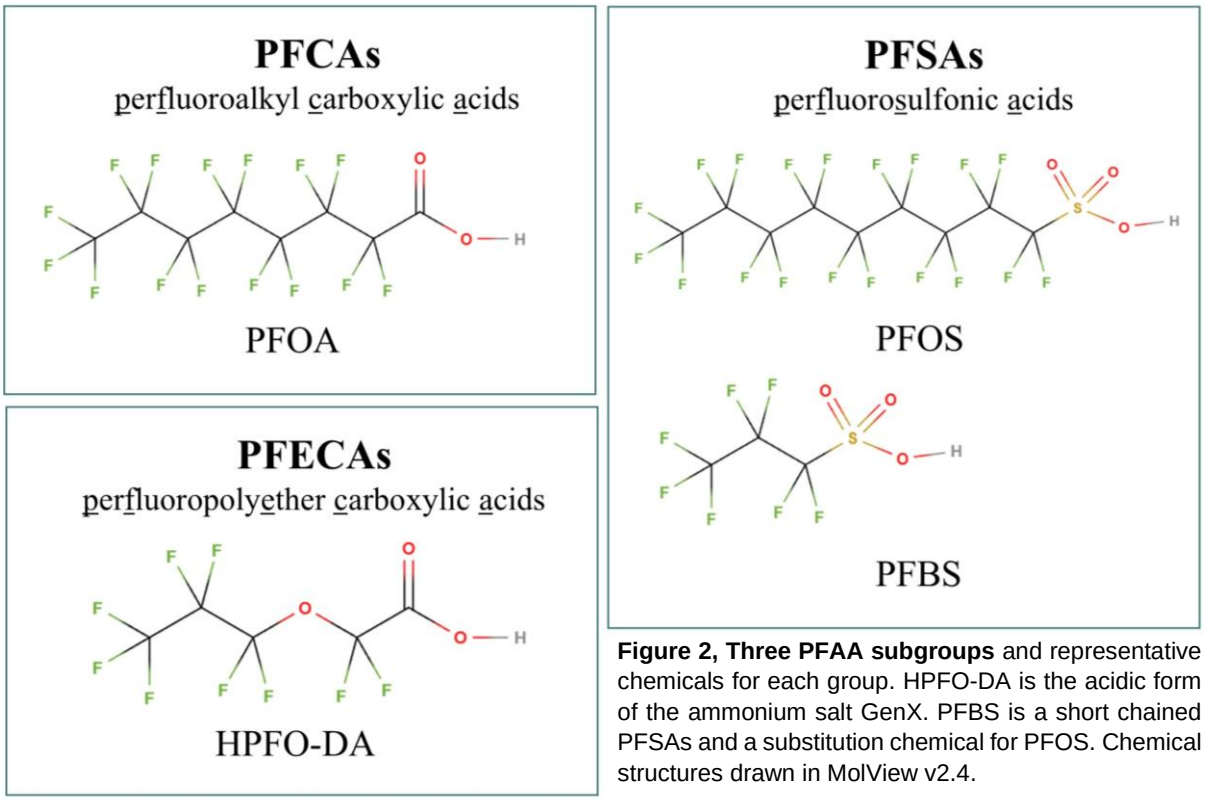

Figure 2, Three PFAA subgroups and representative chemicals for each group. HPFO-DA is the acidic form of the ammonium salt GenX. PFBS is a short chained PFSAs and a substitution chemical for PFOS. Chemical structures drawn in MolView v2.4.

The release of PFAS into the environment can be expected all throughout the compounds life cycle, from the supply chain and production process to use and disposal of the final product (Ahrens & Bundschuh, 2014). The head group of PFCAs and PFSAs are readily deprotonated into their water-soluble dissociated anions which inevitably leads to their accumulation and mobility in aquatic systems (Land et al., 2018; W. Liu et al., 2019; Sunderland et al., 2019). Further, environmental and human exposure has been linked to the widespread use of plastic products and global plastic pollution. PFAS

coatings used for plastic and synthetic textiles are known to desorb both in biota and environmental compartments (Wee & Aris, 2023). This allows for PFAS to make their way into both aquatic and terrestrial food webs where especially long chained compounds will bioconcentrate, bioaccumulate and biomagnify along trophic chains (Conder et al., 2008). Concerns has been raised that microplastics could act as vectors for PFAS, transporting them both long distances and directly into organisms (Salawu et al., 2024; F. Wang et al., 2015).

3.2 Legacy long-chain PFASs

PFOA and PFOS are polyfluorinated acids and have historically been two of the most utilized PFAS globally (Wee & Aris, 2023; US EPA). They are also the most extensively studied PFAAs in an otherwise largely under studied chemical class (Ankley et al., 2021; Glüge et al., 2020; Wee & Aris, 2023). These two long chained PFAAs are the terminal product of biotic and abiotic degradation processes of multiple common types of PFAS precursors (Buck et al., 2011; Kolanczyk et al., 2023; Sunderland et al., 2019). Production and commercial use for PFOA and PFOS started in the mid-20th century, and they were both used freely until the early 2000s when proof of PFOS widespread bioaccumulation in wildlife raised concerns (Giesy & Kannan, 2001; Wee & Aris, 2023). This prompted regulations and stewardship programs that lead to the eventual phase out of the long chained PFAAs by major global manufacturers (Buck et al., 2011; Wee & Aris, 2023).

In year 2000 3M, one of the major PFSA manufacturers in the US, announced a rapid phase out of their PFOS production (Olsen et al., 2008). Following this in 2006 an agreement between the US environmental protection agency (USEPA) and eight major producers was reached and a gradually phase out of long chained PFCAs and their precursor commenced (USEPA, 2006). This was the start of a gradual shift away from the long chained PFAAs. At present practically all use of both PFOA and PFOS is prohibited in the EU, US and Japan (Land et al., 2018; Sunderland et al., 2019). Today both PFOS and PFOA are listed under the persistent organic pollutants (POPs) regulation. PFOS was listed 2009 in Annex B (Restriction) and PFOA 2019 in Annex A (Elimination) (Regulation (EU) 2019/1021).

3.3 Next generation PFAS

Following the phase of the long chained PFAAs a new generation of fluorochemicals has been introduced. The main substitution strategy has been to switch to short chained (<C6), less bio accumulative PFAS with alternative head- and linking groups (Buck et al., 2011; Gomis et al., 2018; Xu et al., 2021). PFBS is a short chained alternative PFSA that has seen an increased use in the wake of the PFOS phase out (Li et al., 2023; W. Liu et al., 2019). GenX is another substitution chemical and was introduced by the chemical company DuPont in 2009 as an alternative to PFOA for use in fluoropolymer manufacturing (EPA, 2021).

GenX is the commercial name of the ammonium salt of hexafluoropropylene oxide dimer acid (HFPO-DA). HFPO-DA is classified as a perfluoropolyether carboxylic acid (PFECAs). PFECAs resemble the structure of PFCAs but have ether linking groups within their carbon chain (see figure 2). The ether linkages insert additional oxygen molecules into the carbon chain, making the molecules more prone to form hydrogen bonds. The potential to form hydrogen bonds in turn increases PFECAs hydrophilicity and decreases the bioaccumulation potential compared to PFCAs (Gomis et al., 2018; Pang et al., 2019). In 2018 GenX was manufactured at 10-100 tons per year in the EU (Mullin et al., 2018). However, in 2019 the inclusion of GenX on ECHA´s Candidate list of substances of very high concern (SVHC) put a halt to the continued use and production of GenX on the European market (ECHA, 2025).

3.4 Environmental presence of PFAS

The discovery of widespread PFAS contamination and subsequent public concern has unfolded in North America and Western Europe in the last two decades (Land et al., 2018; Wee & Aris, 2023). The environmental monitoring efforts have primarily been focused on public drinking water supplies, as high PFAS levels in drinking water have been identified as a major exposure route for the human population (Hu et al., 2016; Sunderland et al., 2019; Vestergren & Cousins, 2009). The US and, notably, Sweden have the highest measured concentrations of both PFOA and PFOS in drinking and surface water.

Record measurements of 522 ng/l PFOA and 2280 ng/l PFOS has been reported in surface waters in Sweden (Wee & Aris, 2023).

According to the Forever Pollution Project´s (2024) compilation of PFAS monitoring data the general combined concentration of PFAS in European freshwater is within 1-100 ng/l. However, hotspots with spiking concentrations (>100 ng/l) are found around point sources like manufacturers of PFAS products, airports and sites used for firefighting training (Norström et al., n.d.). PFOA and PFOS constitutes the overarching part of the identified PFAS contaminants (Dagorn et al., 2024; W. Liu et al., 2019).

Studies of PFAS levels in foodstuff show that consumption of fish and seafood contributes significantly to the human dietary exposure (Augustsson et al., 2021; Barbo et al., 2023). According to monitoring data by USEPA on commercial freshwater fish from US lakes and streams the median total PFAS concentration was 11,800 ng/kg (11.8 µg/kg) for fillets sold between 2013 and 2015. Several locations had measured concentrations of >100,000 ng/kg (>100 µg/kg) (Barbo et al., 2023).

4. Toxicological effects of PFAS on aquatic organisms

Due to their persistence in the environment, long range transport and bioaccumulation potential PFAS are suspected to put toxicological pressure on a wide range of aquatic organisms (Ahrens & Bundschuh, 2014; Buck et al., 2011; J. W. Lee et al., 2020). The persistent organofluorides have been established as driving factors of multiple adverse outcome pathways (AOP) leading to reproductive toxicity, growth and developmental defects, endocrine disruption, neurological defects, hepatoxicity and carcinogenesis in aquatic organisms (Ahrens & Bundschuh, 2014; J. W. Lee et al., 2020). The activation of nuclear receptors, induction of oxidative stress, disruption of immunological systems and direct interactions with membranes are believed to be the main molecular initiating events to subsequent AOPs (J. W. Lee et al., 2020).

4.1 Induction of oxidative stress and antioxidant systems

Oxidative stress is caused by a cellular imbalance between oxidation agents and antioxidants. The increased levels of reactive oxygen species (ROS) connected to PFAA exposure could be linked to interference of fatty acid oxidation, disruption of the mitochondrial membrane´s selective permeability and activation of peroxisome proliferated-activated receptors (PPARs) (J. W. Lee et al., 2020).

Induction of the Acyl-CoA oxidase 1 gene (ACOX1) together with elevated levels of antioxidant proteins following PFAAs exposure suggests that elevated β- oxidation is one inducer of oxidative stress (Arukwe & Mortensen, 2011; J. W. Lee et al., 2020). Acyl-CoA oxidase 1 is the first and rate limiting enzyme of the peroxisomal β-oxidation pathway (Vluggens et al., 2010a). During β- oxidation fatty acids are broken down through a cycle of catabolic reactions, simultaneously reducing the electron transporting molecules NADH and $FADH_2$. These transport molecules supply the mitochondrial respiratory chain with electrons (Berg et al., 2019). The xenobiotic induced β- oxidation thereby causes an elevated electron flow to the mitochondrial electron transport chain.

A side effect of the oxidative phosphorylation in the electron transport chain is leakage of electrons at the protein complexes resulting in the reduction of oxygen into the highly reactive oxygen species superoxide ($O_2^{\cdot-}$). The main defence against ROS mediated oxidative damage is the antioxidant enzyme superoxide dismutase (SOD). Superoxide dismutase exists in two main forms in eukaryotic cells, manganese superoxide dismutase (SOD-Mn) in the mitochondria and the cytosolic copper-zinc superoxide dismutase (SOD-Cu). These antioxidant enzymes rapidly partition the reactive oxygen anion into the stable species O_2 and hydrogen peroxide (H_2O_2) (Berg et al., 2019).

The concentration of H_2O_2 outside of mitochondria is regulated by several antioxidant systems. In the glutathione (GSH) system GSH is oxidized by H_2O_2 into its oxidised form GSSH, thereby eliminating the ROS by its conversion into H_2O. This reduction of H_2O_2 is catalysed by glutathione peroxidase (GPx), and the reverse reaction that restores GSH is catalysed by glutathione reductase (GR) (Berg et al., 2019).

Low levels of ROS production is essential for cellular function as mitochondrial H_2O_2 is a secondary messenger in intracellular signalling and regulates, among other functions, apoptosis, phosphorylation

signalling and activates several nuclear factors, like erythroid 2–related factor 2 (Nrf2) (Averill-Bates, 2024; J. W. Lee et al., 2020). Nrf2 in turn regulates the expression of several antioxidants. Imbalance in the redox environment due to increased ROS production is, however, associated with accelerating aging and other adverse outcomes including tumorigenesis (J. W. Lee et al., 2020).

4.2 Peroxisome proliferator activated receptors (PPARs)

Peroxisome proliferator activated receptors (PPARs) are ligand activated nuclear receptors, involved in a variety of regulatory actions, including the regulation of energy homeostasis. Three subgroups of PPARs are present in eukaryotic cells: PPAR-alpha(α), PPAR-beta(β) and PPAR-gamma(γ). The expression of these isoforms varies across tissue types (N. Wagner & Wagner, 2020).

One example of endogenous PPAR ligands are free fatty acids. Upon activation the PPARs dimerize with the retinoid X receptor (RXR) and thereafter binds, aided either by ligands or on its own, to peroxisome proliferator-responsive elements. The binding regulates the transcription of key enzymes in lipid metabolism, like ACOX1 (Decara et al., 2020; Vluggens et al., 2010b). PPAR-α is a major regulator of fatty acid metabolism in the liver and is active during energy deprived cellular conditions. It inhibits glycolysis while triggering catabolism of fatty acids via upregulation of genes involved in gluconeogenesis and β-oxidation (van Raalte et al., 2004; K.-D. Wagner & Wagner, 2010).

Beyond their endogenous substrates the PPARs have multiple known classes of exogenous ligands. Among them the activation of PPARα and PPARγ has been shown both in human and rat ligand-based reporter assays for multiple PFAS (Evans et al., 2022; Nielsen et al., 2022; Wee & Aris, 2023). Evans et al. (2022) reported HFPO-DA, both as a free acid and an ammonium salt to be potent ligands to PPARα and PPARγ. Arukwe & Mortensen (2011) showed upregulation of the PPARs together with increased ACOX1 expression in liver and kidney tissue of Atlantic salmon fed PFOA spiked food. This finding suggests that the previously mentioned connection between PFAA exposure and β-oxidation could be connected by their PPAR interactions.

4.3 PFAS biomarkers

Altogether, PFAS triggers multifarious responses in aquatic species by disrupting multiple interconnected cellular pathways. As a result, no single, simple biological marker indicative of PFAS-induced toxicity has been established. Instead, multiple suggested markers have been proposed, including genes related to lipid metabolism and oxidative stress.

Traditionally, early indicators of xenobiotic effects are related to the enzymes involved in cellular detoxification pathways. Xenobiotic metabolism is generally divided into three phases: activation, conjugation and elimination (Phang-Lyn & Llerena, 2025). In phase I enzymes introduce or uncover polar functional groups in preparation for the following transformations (Iacopetta et al., 2023). The Cytochrome P450 superfamily (CYP) are major enzymatic actors in phase I and activates its substrates via hydroxylation (McDonnell & Dang, 2013). In phase II target compound are conjugated to a high-energy polar metabolites like glutathione (GSH) by enzymes such as glutathione S-transferases (GSTs). This further increases its hydrophilicity and facilitates its elimination through efflux systems (Aloke et al., 2024). GSTs are also involved in antioxidant defence (Röth et al., 2011).

However, the utility of these biotransformation-related enzymes as biomarkers for PFAS exposure is not well established. Reported CYP responses vary across species, experimental designs and differs between PFAS chemicals (J. W. Lee et al., 2020). PFOA and PFOS are not believed to be bio transformed but are instead the end product of several detoxification pathways (Kolanczyk et al., 2023). Despite the lack of characterized metabolites of PFOA and PFOS, studies suggests that the PFAAs activates the AhR and PXR receptors, thereby influencing enzymes involved in the three phases of biotransformation (Han et al., 2012; Tilton et al., 2008; Wei et al., 2008).

Interestingly, upregulation of CYP activities has been documented for cells cultured in enhanced systems, such as 3D-configurations or in co-culture systems (Krøvel et al., 2011; Lammel et al., 2019). This suggests that the CYP enzymes may be a potential marker not only for xenobiotic responses but potentially also for cellular differentiation driven by altered culturing conditions.

Aim

The aim of this study was to establish a co-culture system combining two rainbow trout cell lines: RTgutGC and RTL-W1. The co-culture setup integrates an *in vitro* gastrointestinal barrier model grown in transwell inserts with RTL-W1 cultured in the well below. The project ultimately seeks to evaluate the usefulness of this co-culture model for studying effects of dietary exposure and sub-lethal effects of environmental contaminants. Additionally, this thesis is part of a broader research project focused on characterizing sub-lethal effects of per- and polyfluoroalkyl substances (PFAS), including both legacy long-chain compounds and newer replacement chemicals. PFAS were therefore chosen as model compounds for this initial testing stage of the co-culture system.

The project was structured around five interconnected parts, each with its own specific aim:

1. Establish the RTgutGC epithelial barrier in transwell inserts to create the foundation for the co-culture system.
2. Screen PFAA cytotoxicity to decipher their relative toxicity rankings and identify suitable sub-lethal exposure concentrations for subsequent co-culture exposure.
3. Asses *in vitro* gene expression responses by comparing transcriptional trends between the co-cultured and monocultured cells.
4. Conduct an *in vivo* feeding experiment on rainbow trout with PFOA-spiked feed and compare liver and intestinal gene expression responses to the *in vitro* data.
5. Compare basal CYP1A1 expressions between RTL-W1 cells and the rainbow trout liver tissue to evaluate xenobiotic metabolic activity across different cell culturing systems.

Methods and materials

1. Chemicals

Perfluoro-2-propoxypropanoic acid (HPFO-DA) (CAS 13252-13-6), Perfluorooctanoic acid (PFOA) (CAS 335-67-1) and Perfluorobutanesulfonic acid **(**PFBS) (CAS 375-73-5) was purchased from Sigma-Aldrich. Cell culture medium phenol red free Leibovitz´s L-15 medium, fetal bovine serum (FBS), penicillin and streptomycin were purchased from Gibco, Thermofisher Scientific, Sweden. Ethylenediaminetetraacetic acid (EDTA), ethanol, methanol, copper sulphate ($CuSO_4$), AlamarBlue and 5-carboxyfluorescein diacetate acetoxy methyl ester (CFDA-AM) solutions were purchased from ThermoFischer Scientific, Sweden. L-15ex minimal medium buffer was prepared from three pre-prepared stock solutions. Stock A: NaCl, KCl, MgSO4x7H2O, MgCl2x6H2O, Solution B: $CaCl_2$ x $2H_2O$, Solution C: Na_2HPO_4 x $12H_2O$, KH_2PO_4, Solution D: Sodium pyruvate (Sigma P5280), Solution E: Galactose (Sigma G5388).

2. Preparation of exposure solutions

Stock solutions of the PFAAs were prepared in methanol (10 mg/ml) and kept in glass vials in the fridge. L-15ex stock was prepared the day prior to exposure by evaporating the methanol stock under N_2 gas. L-15ex was thereafter added to the scintillation vial and they were shaken over night at 450 rpm. Further dilutions of the L-15ex stock were made the morning of the exposure.

3. Cell lines

Two cell lines where used: RTgutGC and RTL-W1, epithelial intestinal and liver cells respectively, from rainbow trout (*Onchorhynchus mykiss*). RTgut-GC and RTL-W1 both stem from spontaneously immortalised cells from non-tumorous tissue. The cells grow adherent to a substrate in a monolayer of epithelial-like cells (Kawano et al., 2011; L. E. Lee et al., 1993). Cells were kindly donated by Professor K. Schirmer at EAWAG, Switzerland.

3.1 Routine cell culturing

RTgutGC and RTL-W1 were continuously cultured in basal, phenol red-free Leibovitz´s L-15 medium supplemented with 5%(v/v) fetal bovine serum (FBS). The cultures were grown in 75 cm^2 culture flasks (TC T75 flasks, Sarstedt) at 19 °C, normal atmosphere. The cultures were split routinely (1:2 or 1:3) when confluent. Detachment was routinely done with 0.2 g/l ethylenediaminetetraacetic acid (EDTA)/phosphate buffer saline (PBS) (Sigma-Aldrich) treatment for 3 minutes followed by 0.25% trypsin-EDTA solution until cells lifted. A Büchner chamber was used to determine the cell count.

4. Cytotoxicity

4.1 The Assays

The cytotoxicity screenings were done according to a modified protocol based on OECD Test No. 249 (OECD, 2021). The cytotoxicity readings provide two cell viability measures via two fluorescent probes: AlamarBlue and CFDA-AM. AlamarBlue indicated loss of metabolic capacity while the CFDA-AM assay gives a response that could indicate loss of membrane integrity, loss of esterase activity or a combination of both outcomes (Rampersad, 2012; Schreer et al., 2005).

The AlamarBlue assay is based on the redox indicator resazurin and enables quantitative measurements of cell viability. Resazurin is a blue, non-toxic, membrane permeable and weakly fluorescent compound. This reagent can be added directly to cell cultures and will be reduced into the highly fluorescent red compound resorufin in viable cells. The change in colour and fluorescent intensity of the reduction product can be used as a proxy for cell viability (Rampersad, 2012).

The 5-CFDA-AM assay is used to quantify viability by measuring enzymatic activity and membrane integrity. The probe 5-CFDA-AM is a non-fluorescent and membrane permeable esterase substrate. Esterases will cleave the acetoxymethyl group (AM) in viable cells, thereby producing the highly fluorescent and electronically charged 5-carboxyfluorescein (5-CFDA). The assay was previously believed to solely indicate loss of membrane integrity as the fluorescent and charged compound will

leak out of a disrupted membrane. However, the fluorescent intensity depends on both the esterase activity and the membrane integrity (Schreer et al., 2005).

4.2 Cytotoxicity protocol

4.2.1 Seeding of cells

Cells were detached from confluent T75 flasks and cell suspensions were prepared in serum supplemented L-15 medium (5% FBS). The cells were seeded in 96-well plates (TC, Cell +, F, Sarstedt) 24 hours prior to exposure at 350,000 cells/ml, 100µl per well (=35,000 cells/well). The outermost wells of the plates, creating the outer frame, were not seeded but instead filled with 100 µl with L-15ex /well.

4.2.2 Exposure

Dilution series of the PFAAs were prepared in L-15ex. The culturing medium was exchanged with the exposure solutions and the plates were incubated for 48 hours at 19 °C. A five-concentrations dilution series of $CuSO_4$, starting at 70 mg/l, with a dilution factor of 2.5, was run in each plate as a positive control.

4.2.3 AlamarBlue & CFDA-AM assay

After 48 hours exposure incubation the cells were carefully washed once with 150 µl of L-15ex/well. One joint probe solution with AlamarBlue (1.25% v/v) and CFDA-AM (4µM) was prepared in L-15ex. The solution was kept in an opaque container, away from light sources. 100 µl/well of the probe solution was added to all seeded wells and six additional wells from the non-seeded frame. The non-seeded wells were used for no-viability blank measures. The plates were covered and incubated at 19°C for 30 minutes. A Spectra MAX Gemini EM plate reader (Molecular devices) was used for the fluorescent measurements. The fluorescent intensity was measured simultaneously at the excitation/emission wavelengths at 523/590 nm for the AlamarBlue assay and 485/535 nm for the CFDA-AM assay.

4.3 Data treatment

4.3.1 Viability calculations

The viability was calculated by first subtracting the average no-viability measurements from the other data. The average fluorescent unit (FU) of the controls (no-exposure) was calculated and was thereafter used to determine the viability percentage for the exposed cells with the following formula:

$$Viability\ (\%) = \frac{Exposed\ cells\ (FU)}{Average\ controls\ (FU)} \cdot 100$$

4.3.2 Data presentation & analysis

Initial data treatment and organisation was carried out in Microsoft Excel. The excel sheets were imported into RStudio/2024.09.1+394 (“Cranberry Hibiscus”). Further treatment and plotting were performed with the ggplot2 and drc packages. Scatterplots were made with ggplot2 and regression analysis with the drc package. For the initial range finding a simple LOESS (locally estimated scatterplot smoothing) was performed using the loess_fit function.

The drm (dose response model) tool was used to fit a non-linear dose response model to the refined dose response curves. 4-parameters log-logistic and Weibull model (W1.4) were fitted. The fit was evaluated based on the AIC (Akaike Information Criterion) that weights the number of parameters (complexity of model) vs the fit. Weibull (W1.4) was determined to be the best fit and the EC50 values corresponding to this model were extracted.

5. Co-culture setup

5.1 Seeding of RTgutGC on transwell inserts

Transparent ThinCerts™ inserts (4.52 cm^2 culturing area, pore size 3.0 µm) were placed in 6-well culture plates (TC, Cell +, F, Sarstedt) (9.6 cm^2/well). RTgutGC cells were seeded in the inserts at 77.000 cells/cm^2 in serum free L-15 medium supplemented with 1% penicillin/streptomycin (PS). The wells beneath the inserts were filled with 3.5 mL of serum and antibiotic supplemented L-15 medium (5% FBS, 1% PS) for the duration of the gut cell barrier establishment. A schematic of the setup is pictured

below in figure 3. Plates were sealed and incubated at 19°C, normal atmosphere. Culture mediums were exchanged weekly.

Figure 3. Setup for RTgutGC barrier establishment. RTgutGC grown on transwell inserts. Inserts were placed in the well of a 6-well plate. Insert containers filled with L-15 medium, wells filled with serum supplemented L-15 medium. PS added to combat potential bacterial contamination.

5.2 Trans epithelial electrical resistance (TEER) measurements

Trans epithelial electrical resistance (TEER) was used to track the progression of the barrier formation on the permeable membranes. An epithelial Voltohmmeter EVOM^2 (WPI) equipped with a STX2 "chopstick" electrode (WPI) was used for the TEER measurements. The electrodes were placed at either side of the insert wall, one electrode in the well and one in the inserts. An alternating current voltage signal was applied, the resulting current measured and the Voltohmmeter converted the signal to corresponding resistance based on Ohm´s law (Srinivasan et al., 2015).

Measurements were taken in a separate reference plate, seeded and treated identically to the experimental plates. The electrode was moved around the insert so that 10 different readings were recorded per inserts. TEER was measured at day 7-, 14- and 21-days post seeding. Culture medium was changed weekly in both reference and experimental plates, coinciding with the TEER measurements.

In the reference plates three inserts were seeded with gut cells and two inserts were left empty for use as blank references. TEER was calculated by subtracting the blank measurements from the values in the seeded inserts and presented as Ω per cm^2 of insert area (Ω x cm^2), as described below.

$$TEER\ (\Omega cm^2)\ = (TEER\ seeded\ insert\ (\Omega) - TEER\ cell\ free\ insert(\Omega)) \cdot insert\ area\ (cm^2)$$

5.3 Co-culture assembly and exposure procedure

Exposure solutions (control, low, high) were prepared in L-15ex the day prior to exposure and left over night on a shaking table (450rpm). RTL-W1 cells from confluent T75 flask were split 1:2 and seeded in 6-well plates 24 hours prior to exposure.

The final co-culture set up was assembled on the day of exposure. The wells of the RTL-W1 plates were emptied of culture medium. The inserts were emptied of the growth medium, re-filled with 3.5 ml of exposure solution, removed from their original plate and placed on top of the liver cells. Finally, 3.5 ml of L-15 medium (5% FBS, 1% PS) was added to the RTL-W1 cells in the wells. Exposure set-up pictured below in figure 4. The exposed co-culture was sealed and incubated for 48 hours at 19°C.

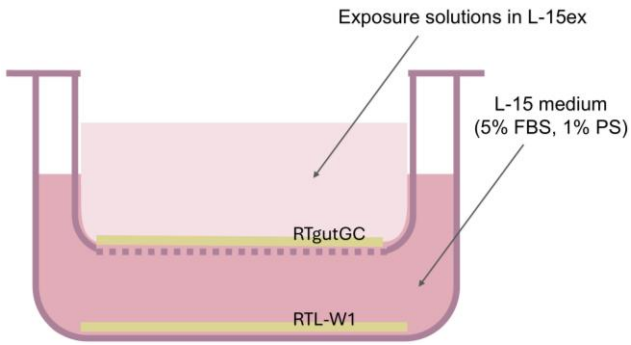

Figure 4. Setup for RTgutGC/RTL-W1 co-culture. RTgutGC barrier in transwell inserts placed in well with RTL-W1 seeded below. Insert container filled with L-15ex exposure solution, wells filled with serum supplemented L-15 medium.

5.4 Monoculture setup

RTL-W1 was seeded (1:2) in 6-well plates (TC, Cell +, F, Sarstedt) in L-15 (5% FBS) 24 hours prior to exposure. The culture medium was removed and exposure solutions (control, low or high) in L-15ex were added. The plates were sealed and incubated for 48 hours at 19°C.

For the investigation of medium induced effect on transcription responses an additional experiment was setup. RTL-W1 was seeded (1:2) in a 6 well plate in L-15 (5% FBS). 24-hours post seeding the medium was exchanged either for fresh L-15 (5% FBS) or L-15ex. The plates were sealed and incubated for 48 hours at 19°C and the cells were thereafter harvested as described below in section 5.5.

5.5 Cell harvesting

At the end of the 48-hour exposure the medium in both the inserts and wells were collected for future chemical analysis. The inserts were removed to an empty plate where they were treated with EDTA /PBS (1ml/ insert) for 5 min. The EDTA/PBS was removed, and trypsin (200 µl/insert) was added. The trypsin treatment was halted after 3 min by addition of L-15 (5% FBS) medium (1 ml/insert). The cell suspensions were removed to Eppendorf tubes.

The cells in the wells were treated with EDTA/PBS (2ml/ well) for 5 min. The EDTA/PBS was removed, and trypsin (300µl/well) was added. After 3 min L-15 (5% FBS) was added to collect the cells and the cell suspension was removed to Eppendorf tubes.

The detached cells in the Eppendorf tubes were pelleted by centrifugation (300g, 5 min). The supernatant was removed, and the pellet was disrupted by flicking the tube prior to resuspension in 350 µl of Qiagen´s RLT buffer plus with 40 mM dithiothreitol (DTT). The cell solutions were removed to a Qiashredder Mini Spin Column and spun down by centrifugation (17,000g, 2 min). The flow through was rapidly frozen on dry-ice and stored at -80°C.

6. RNA extraction

Cell material

The cell lysates were allowed to thaw at room temperature and thereafter added directly to gDNA Eliminator spin columns. The RNA extraction continued following the RNeasy Mini Kit protocol.

Tissue material

Tissue from liver and intestine were collected from rainbow trout during the PFOA feeding experiment described below in method section 9. The tissue was kept on dry ice during handling. 10 mg pieces were carved from the collected tissue. The pieces were added directly into Eppendorf tubes prepared with RLT buffer plus and one stainless steel bead/tube (5mm) (Qiagen). The tissue was homogenized in a TissueLyser II (Qiagen) for 2x2 min at frequency 25/s. The lysates were thereafter centrifuged (17,000g, 3 min). The supernatant was removed to a gDNA Eliminator spin column. The RNA extraction continued according to the RNeasy Mini Kit protocol´s instructions recommended for tissue samples.

Measuring of RNA concentrations

RNA concentrations were measured using Thermo Fisher Scientific NANODROP ONE. Additionally, adequate nucleic acid quality and RNA concentrations were confirmed on a random selection of samples using tape station RNA kit (Aqilent Technologies), and the 2200 tapestation reader (Agilent Technologies). Tape station data found in appendix 2. The RNA samples were stored at -80°C.

7. cDNA synthesis

iScript cDNA Synthesis Kit was used for the cDNA synthesis. All samples were diluted in nuclease-free water to a uniform final concentration (400ng/15 µl for cells and 1000ng/15 µl for tissue). To 15 µl of diluted samples 5 µl of mastermix (4µl 5x iScript Reaction Mix and 1µl iScript Reverse Transcriptase per sample) was added. Controls for residual gDNA were created on a random selection of samples by excluding reverse transcriptase. The samples were incubated in a BIORAD MyCycler™ Thermal Cycler running the following protocol (table 1):

Table 1, thermal cycle protocol for cDNA synthesis.

Step	Time	Temperature
Priming	5 min	25 °C
Reverse transcription	20 min	46 °C
RT inactivation	1 min	4 °C

8. Quantitative real time PCR (qPCR)

cDNA was diluted 1:20 in nuclease free water. A master mix was prepared with a primer pair (reverse and forward) and SSOSYBR Green (0.50 µl reverse primer + 0.50 µl forwards primer, 5 µl of SSOSYBR Green)/sample). Primers listed in appendix 1. In a 96-well PCR plate (thin wall, BIORAD laboratories) 4 µl of template cDNA and 6 µl of the master mix were added to each well. 3 technical replicates were run for each cDNA sample. No template controls (nuclease free water) and controls for genomic DNA (no reverse transcriptase) were included on each plate. The plates were sealed (Microseal "B" foil seals), centrifuged (1 min, 160 RCF) and run in BIORAD CFX96™ Real-Time System Thermal Cycler using the following two-step RT-qPCR protocol (table 2), followed by a melt curve program (table 3):

Table 2, protocol run on BIORAD CFX96™ Real-Time System Thermal Cycler.

	Initial denaturing	Denaturing template	Annealing & extension
Time	3 min	10 s	30s
Temp	95°C	95 °C	60 °C
Repeat	Once (first cycle)	40 cycles	40 cycles

Table 3, melt curve protocol run on BIORAD CFX96™ Real-Time System Thermal Cycler after the qPCR protocol described in table 2.

Step	Time	Temperature	Repeats
Start	5s	65 °C	Once (first cycle)
Gradual temperature increase		+0.5 °C/ cycle	Until 95 °C (60 cycles)

8.2 Data treatment

The qPCR data was normalized against the relative expression of the house keeping genes Ubiquitin and Actin. The average Cycle threshold (Ct) values for Ubiquitin and Actin were subtracted from the Ct´s of the target genes and the normalised data was transformed ($2^{\wedge -\text{normalised data}}$) to enable the creation of more intuitive plots.

8.3 Statistics

Statistical tests were conducted using IBM SPSS statistics version 30.0. Levene´s test of homogeneity of variance was performed to ensure equal variance across treatment groups. Additionally, Shapiro-Wilk test of normality was performed. For non-significant samples ($p>0.05$) equal variance and normal distribution was confirmed and the adequacy of a parametric analysis of variance was assumed. For these samples a one factor ANOVA was followed by Dunnett´s two-sided ad hoc test.

For samples that did not pass Levene´s - or the Shapiro Wilk test a non-parametric Kruskal Wallis test was performed in addition to the parametric ANOVA. The Kruskal Wallis test was followed by pairwise Mann Whitney U tests. For no samples the non-parametric tests concluded a different outcome (significant/non-significant) compared to the parametric tests. Based on both the parametric or non-parametric tests the null hypothesis (no difference in variance between and within treatment groups) was either rejected or accepted. Level of statistical significance was set to $p < 0.05$.

Additional ranking of significant levels was based on the parametric tests for the samples that passed the Levene´s and Shapiro-Wilk tests. For the other samples the presented significance level was instead based on the non-parametric Mann Whitney U-test.

9. Rainbow trout feeding experiment

Juvenile rainbow trout (150 g ± 50 g) were bought from Vänneån fiskodling AB and kept in circulating dechlorinated tap water (11-12 ºC) in a 12:12 light/dark cycle. The fish were housed together in holding tanks and moved to 300 Liter aquariums 1.5 weeks prior to exposure. In the aquariums the fish were kept alone and introduced to the gelatinous-worm food. Experiments conducted according to ethical permit: Dnr 5.8.18-05185/2024

9.1 Preparation of food for PFAS exposure

Gelatinous worm-based food was created for the PFOA exposure. Three sets of food were created. One control batch without PFOA, one for dosing of 1 mg PFOA/kg fish and 10 mg PFOA/kg fish. One food cube (1 cm^3) constituted the complete dose. Methanol stock was evaporated under N_2 gas, distilled water was added and the PFOA was left to dissolve overnight (450rpm). The PFOA solutions were made 2x the target concentrations (0.15 mg/ml or 1.5 mg/ml).

Frozen red mosquito worms (Blood worms) (AkvarieTeknik) were left to thaw at room temperature and then blended with an immersion blender until homogenous. 0.3 g/ml of gelatine powder was mixed in the PFOA solutions. The PFOA/gelatine mixtures were warmed in a water bath (60°C) until a homogenous gelatine solution had been reached (~5 min). The PFOA/gelatine solutions were mixed 1:1 with the worms. The final mixture was quickly poured into silicone ice cube moulding trays (pictured in figure 5) and excess mixture was smoothed out using a metal spatula. The mixture was left to set at room temperature (~15 min) and then stored in the fridge.

Figure 5. Picture of silicone mould used to create 1 cm^3 cubes of gelatinous worm food packages. Trays originally intended for mini-ice cubes. The gelatinous food cubes were easily detached from the moulds. Moulds from DAY, ordered from: https://www.netonnet.se/art/hem-fritid/koksredskap/day-isform-silikon-mini-2-pack

9.2 Feeding

The food cubes were prepared the morning of the feeding according to the procedure described above. One cube was feed per fish. The fish that ate their cube did so within 15 minutes. Two rounds of feeding were conducted for two batches of fish. After the first feeding and sampling, fish from the holding tank was moved into the tank for acclimatisation and once they were calm and ready the new fish were feed PFOA spiked food.

9.3 Sampling

48 hours post exposure the fish were euthanized by a sharp blow to their head. The fish were dissected, and liver and gut samples collected. The liver was cut into aliquots, and the intestinal samples were collected by scraping of the epithelial cells from the mucosa with a glass slide. The tissue samples were stored in liquid nitrogen until further RNA-extraction, cDNA synthesis and qPCR (described above in section 6,7 and 8).

Results

The following section summarises the key findings of the establishment and gene expression analysis of the *in vitro* gastrointestinal barrier model. Results are presented in five sections: (1) validation of the establishment of an RTgutGC intestinal barrier, (2) cytotoxicity range screening for perfluoroalkyl acids (PFAAs), (3) gene expression analysis following GenX and PFOA exposure for the *in vitro* model, (4) *in vivo* gene expression for rainbow trout fed PFOA spiked food, and lastly (5) comparison of *in vitro* models and *in vivo* data for the classic biomarker CYP1A1.

1. Development of a gastrointestinal barrier on transwell inserts

Transepithelial electrical resistance (TEER) was measured weekly at 7-, 14- and 21-days post seeding. The TEER measurements allowed tracking of the barrier formation and was the main indicator of a successful establishment of a gut cell barrier. A total of four successful insert trials were conducted. For all four trials a stepwise increase in TEER was recorded across three weeks. The average TEER values for the four successful trials are plotted in figure 6, based on the data presented in table 4.

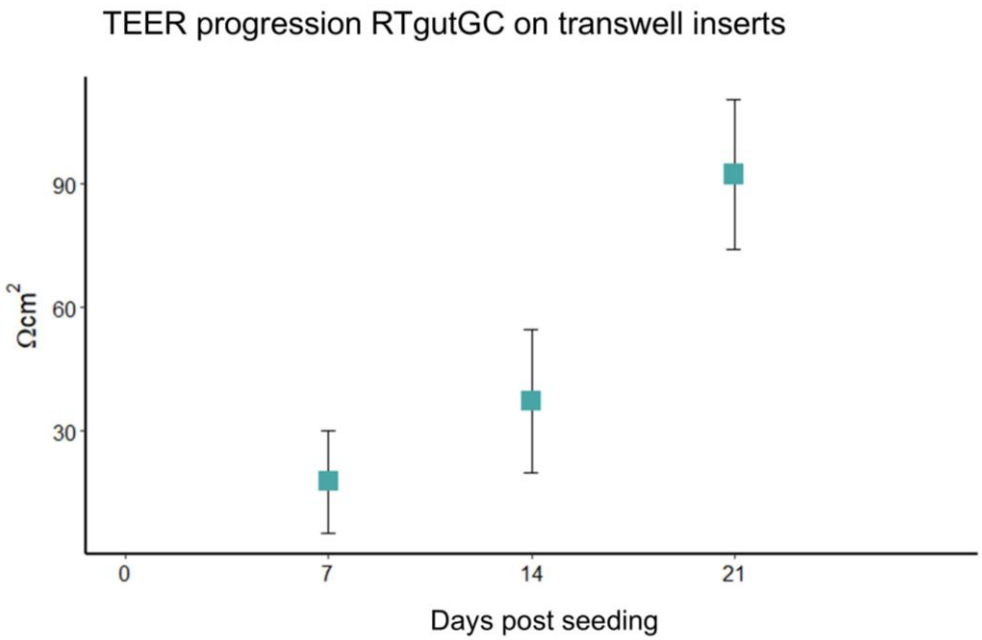

Figure 6. Progression of TEER for RTgutGC as Ω x cm^2 over 21 days. Gut cells seeded in 452 mm^2, 3.0 µm transwell inserts. Each datapoint represents an average of four biological replicates with different passage numbers of RTgutGC. The standard deviation presented as error bars.

The epithelial resistance increased weekly and reach 97 ±18 Ωcm^2 on day 21. One measuring plate was created per trial with seeded inserts allocated for TEER measurements. A sizeable variance in TEER was recorded across both separate trials (n=4) and replicate inserts (n=3) within each trial. A stepwise, accelerating TEER progression was, however, noted for all trials. The standard deviation (SD) within each measuring plate decreased percentagewise as the TEER progressed for all attempts (see table 4).

Table 4, TEER data for four separate trials of RTgutGC seeded on transwell inserts. Progression of TEER for each attempt as the mean value of three inserts and SD for each trial and time point. Percentage of error compared to mean TEER presented in the parentheses. Average TEER, summarising all trials for each timepoint at the bottom.

Trial (#)	TEER ($\Omega cm^{2)}$ day 7	TEER ($\Omega cm^{2)}$ day 14	TEER ($\Omega cm^{2)}$ day 21
1	7.6 ±12.6 (166%)	14.8 ± 7.9 (53%)	69.9 ± 11.1 (16%)
2	19.6 ± 6.4 (32%)	38.6 ±5.8 (15%)	95.0 ± 7.9 (8%)
3	16.9 ± 6.5 (38%)	56.4 ± 6.3 (11%)	115.1 ± 9.4 (8%)
4	26.9 ± 17.8 (66%)	39.3 ± 14.8 (38%)	89.0 ± 12.4 (14%)
Average:	17 ±12	37 ±17	97 ±18

2. Cytotoxicity screening

2.1 Initial range finding

Range finding for PFOA and the two substitution PFAAs GenX and PFBS was conducted with RTgutGC. The 48-hour exposure time was chosen to correspond with subsequent co-culture experiments. The concentration ranges were chosen to be as broad as possible, while not exceeding solubility limits and accounting for the availability of stock. Analysed together the two viability assays gave a preliminary ranking of cytotoxicity among the tested PFAAs. Additionally, the range finding indicated appropriate exposure ranges for investigations of sub-lethal effects for subsequent exposure of *in vitro* models. The results of both viability assays for PFOA, GenX and PFBS are plotted below in figure 7.

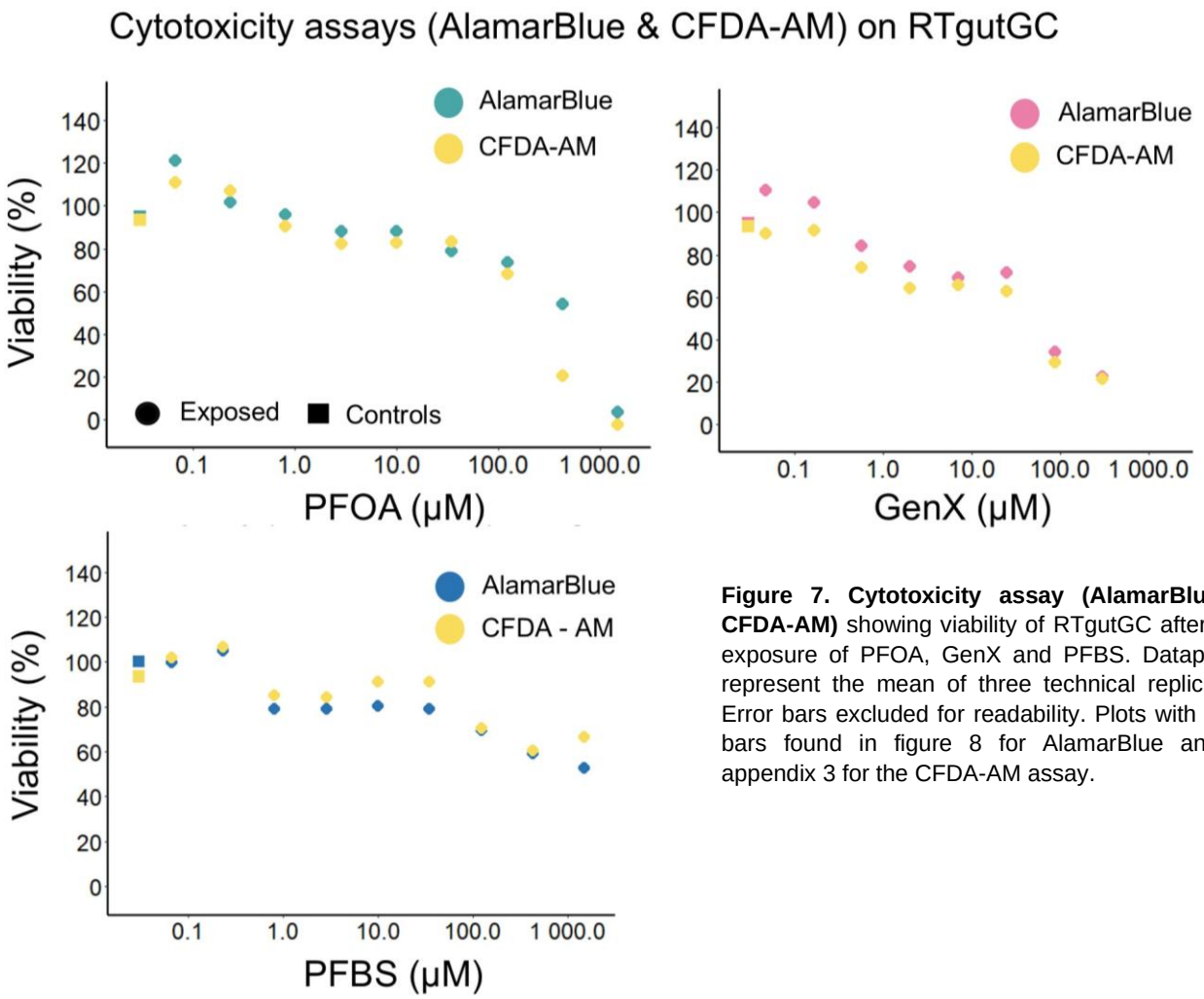

Figure 7. Cytotoxicity assay (AlamarBlue & CFDA-AM) showing viability of RTgutGC after 48h exposure of PFOA, GenX and PFBS. Datapoints represent the mean of three technical replicates. Error bars excluded for readability. Plots with error bars found in figure 8 for AlamarBlue and in appendix 3 for the CFDA-AM assay.

The two cytotoxicity assays showed no indication of differing sensitivity between the two viability endpoints, instead the AlamarBlue and CFDA-AM results closely mirrored each other. In figure 8 the AlamarBlue results are compared between the three PFAAs and overlayed to decipher cytotoxicity trends.

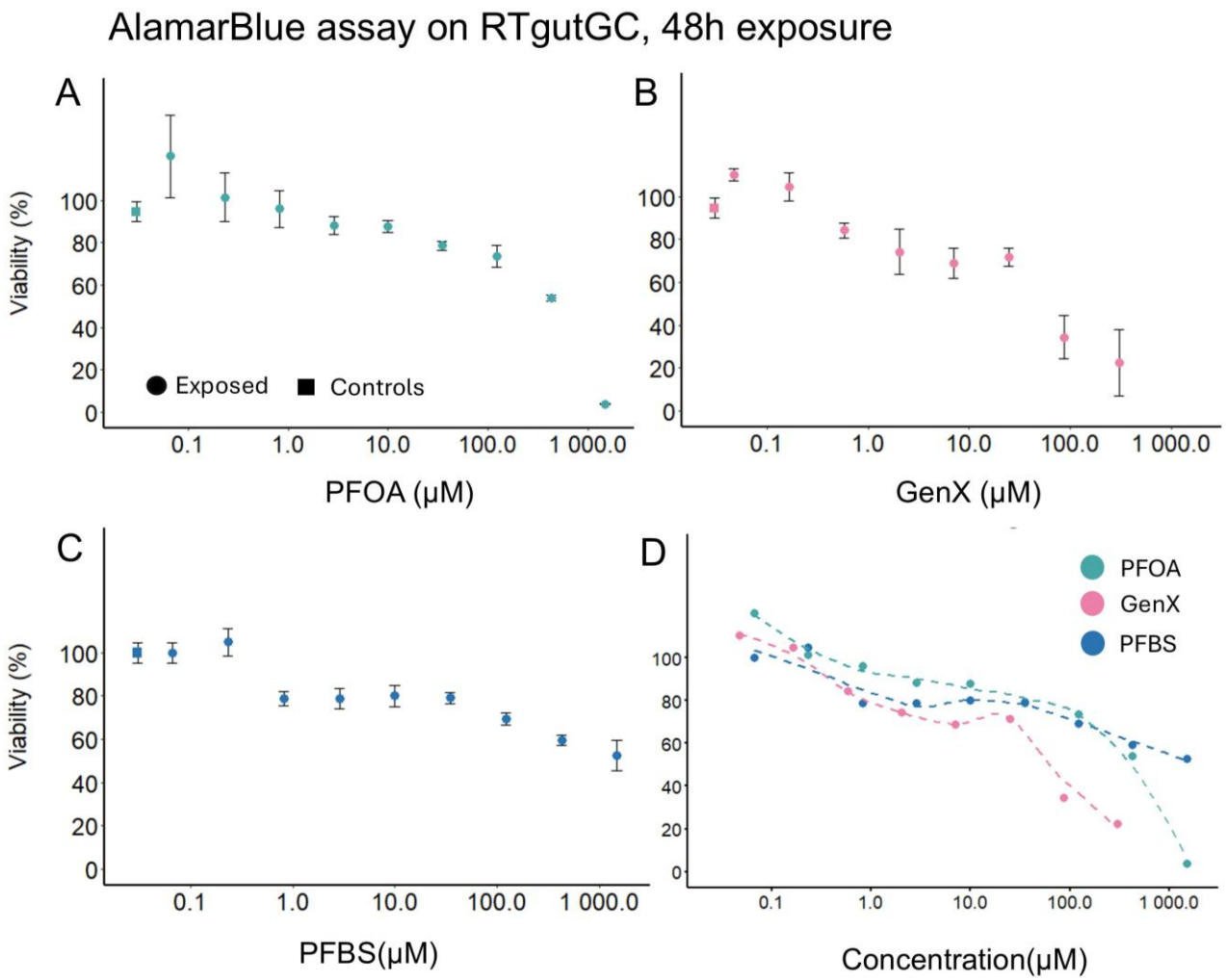

Figure 8, cytotoxicity range finding for PFOA, GenX and PFBS. AlamarBlue cytotoxicity assay performed on RTgutGC in L-15ex medium, 48h exposure. Data points represent the mean of three technical repeats, SD shown as error bars. All dilution series with dilution factor (df) 3.5. **A.** PFOA exposure in range 1500 to $1.9*10^{-2}$ µM. Plotted data representative for three plates with cells of different passage numbers. **B.** GenX exposure range 303 to $4*10^{-2}$ µM **C.** PFOA exposure range 1500 µM to $1.9*10^{-2}$. **D.** PFOA, GenX and PFBS exposure data combined. Local polynomial regression lines (dotted lines) with LOESS (locally estimated scatterplot smoothing) fitted to each data set to simplify visualization of data trends.

Based on the visuals of the simple regression lines fitted with LOESS the preliminary cytotoxicity ranking of the PFAAs is: GenX > PFOA > PFBS. PFOA and PFBS were tested in the same broad exposure range. For PFOA a complete story from full viability to zero viability was captured within the tested range. PFBS exhibited lower cytotoxicity with a minimum viability at 53% for the highest tested concentration (1500 µM). Due to limited availability of GenX the maximum tested concentration was 303 µM. At 303 µM the recorded viability for RTgutGC with the AlamarBlue assay was 23%. CFDA-AM results for the PFAAs plotted with error bars are found in appendix 2.

2.2 Refined range finding on RTgut and RTL-W1

Following the broad screening of the three PFAAs a more refined cytotoxicity test was performed for PFOA on both RTL-W1 and RTgutGC. The lowest concentrations were excluded from the test range and a dilution factor of 2 was chosen for more closely spaced data points. The goal was to investigate

possible differences in sensitivity between the two cell lines and obtain more detailed data in the range of toxicity to enable EC50 calculations. The results of the two assays for both liver and gut cells are plotted below in figure 9 and 10.

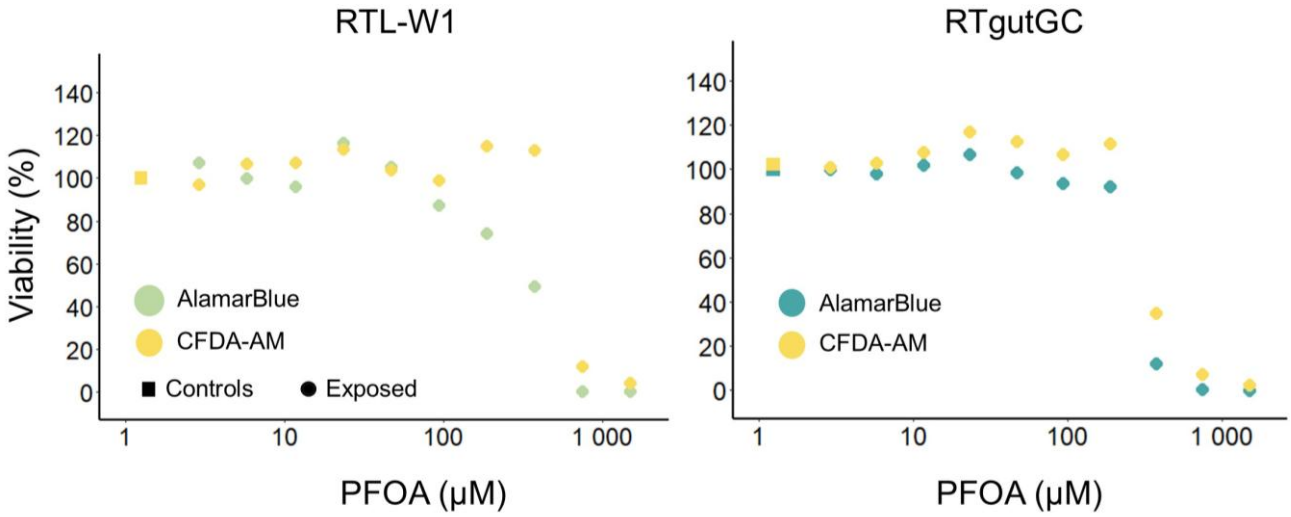

Figure 9. Cytotoxicity assays (AlamarBlue & CFDA-AM) on RTL and RTgutGC. Datapoint represent the mean of three technical replicates, error bars excluded for improved readability. PFOA exposure in concentration range 1500 µM to 2.92 µM, df=2. RTgutGC plot is representative for two plates.

Based on the cytotoxicity screening lethal effects are recorded within the same order of magnitude for the two cell lines. For RTgutGC a steep loss of viability is seen between tested concentrations 188 µM and 375 µM while the AlamarBlue assay shows a more gradual decline in viability for RTL-W1 across a wider concentration range. Notably, CFDA-AM results for RTL-W1 deviates somewhat from the AlamarBlue trend. Based on this AlamarBlue appears to be a more sensitive viability endpoint for RTL-W1. The AlamarBlue data for RTgut and RTL-W1 is overlayed in figure 10.

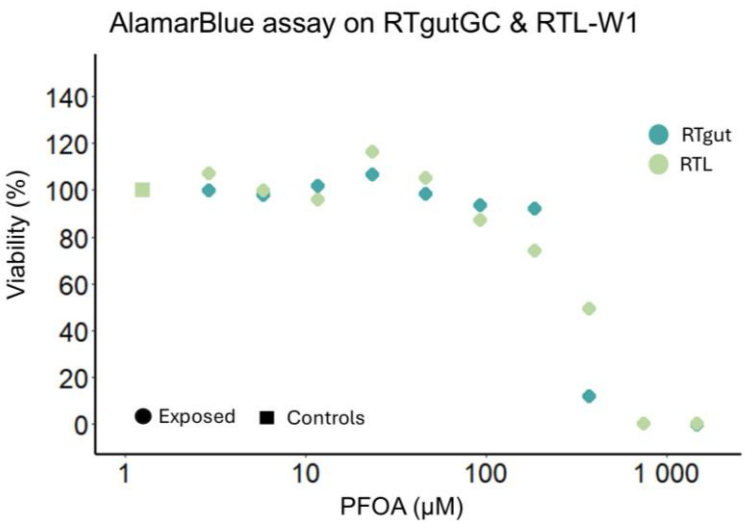

Figure 10. AlamarBlue assay on RTgut and RTL. 48h PFOA exposure in concentration range 1500 µM to 2.92 µM, df=2. Datapoints represents the mean of three technical replicates, error bars excluded for improved readability.

A Weibull model was fitted to both the RTgutGC and RTL-W1 data and effect concentration 50% (EC50) was extracted for each cell line, results displayed in figure 11.

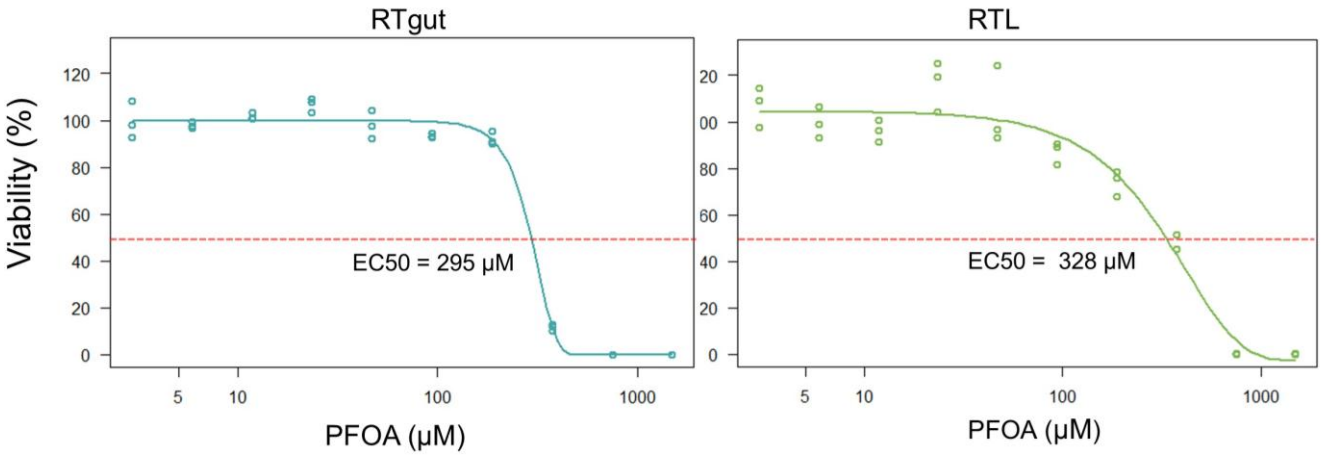

Figure 11. Regression analysis for the dose response curves for RTgutGC (left) and RTL-W1 (right). EC50 for PFOA calculated based on the resulting Weibull model.

Based on the EC50 values for the fitted Weibull model RTgutGC is marginally more sensitive than RTL-W1. However, the steep loss of viability and the failure to capture datapoints within the slope section of the plot is notable for RTgutGC. The difference in slope shape is evident when comparing EC10 values obtained from the Weibull function. The predicted EC10 is 97 µM for RTL-W1 and 197 µM for RTgutGC.

3.Gene expression analysis on the *in vitro* model

The RTgutGC inserts were assembled with RTL-W1 and the complete co-culture setup was then exposed to the PFAAs at two concentrations, one lower and one higher. After 48 hours of exposure the gut cells in the inserts and the liver cells were harvested and the gene expression of a range of genes related to oxidative stress responses, cellular metabolism and biotransformation of xenobiotics were investigated with qPCR (complete list of primers in appendix 1).

3.1 GenX trial

Following the initial cytotoxicity screening, the apparent high toxicity of GenX compared to the other tested PFAAs made it an interesting candidate for further co-culture trials. The exposure concentrations were chosen based on the result of the initial cytotoxicity screening and by referencing concentrations used by Solan, Senthilkumar, et al. (2022) for their PFAA screening on human cell lines. The first successfully established co-culture setup was exposed to GenX at 0.1 µM and 10 µM. The setup was arranged and exposed on day 27 post seeding. Due to doubts about the cell growth progression the FBS concentration was doubled (from 5% to 10%) one week prior to exposure. 10% FBS was also used in the basolateral compartment during exposure.

The expression of oxidative stress related genes showed no significant upregulation (see figure 12). The average expression levels did, however, show a similar trend across the three tested antioxidant enzymes. A slight upregulation at 0.1 µM GenX compared to the control was observed for the three antioxidant genes in the RTgutGC-barrier but could not be confirmed statistically (figure12).

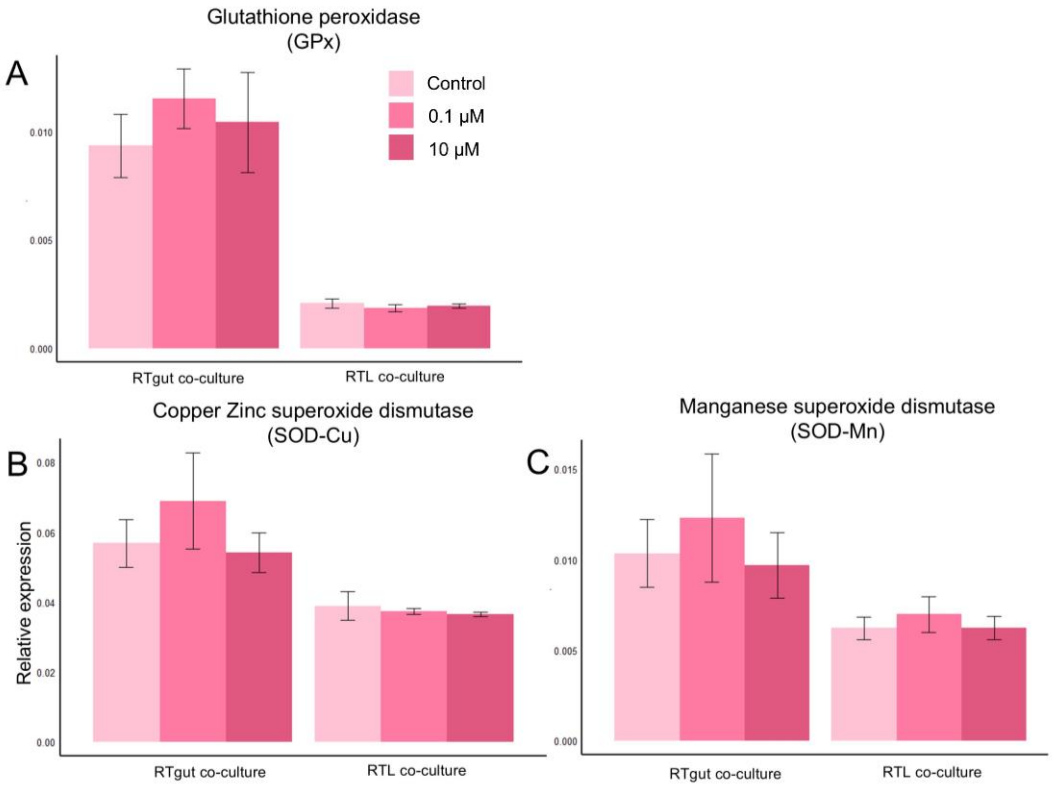

Figure 12. Relative expression of three oxidative stress related genes post GenX exposure. Bars represent the mean value of four or three replicate samples, SD shown as error bars. **A**. Basal GPx expression was significantly higher in RTgutGC compared to RTL-W1. No significant changes in expression were observed but the results suggest an upregulation at 0.1 µM GenX **B.** SOD-Cu showed no significant changes in expression following GenX but a slight upregulation at 0.1 µM was observed, similar to the GPx expression. **C.** SOD-Mn expression closely mirrored the expression of SOD-Cu.

Three genes related to lipid metabolism were tested, two PPARs (α and β) and ACOX1. Large variance in expression across the replicates was recorded for the metabolic genes. No clear trends could be distinguished (see figure 13).

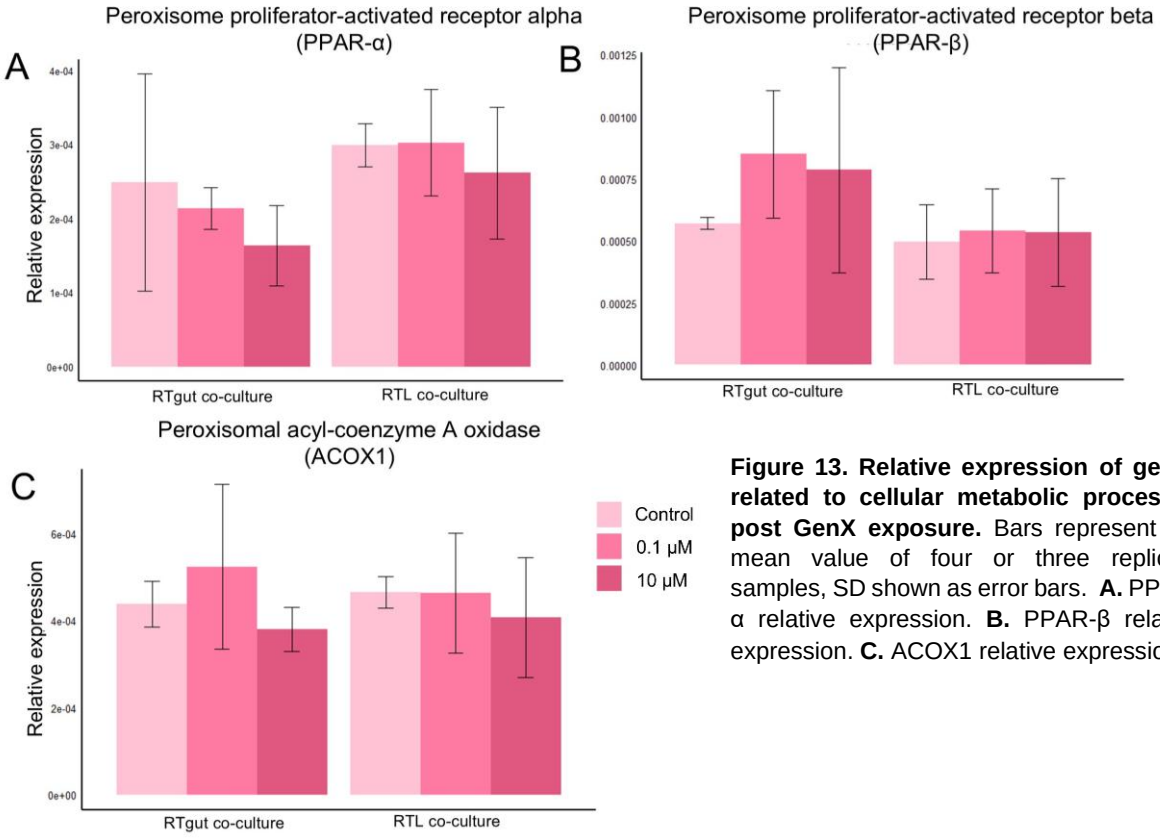

Figure 13. Relative expression of genes related to cellular metabolic processes post GenX exposure. Bars represent the mean value of four or three replicate samples, SD shown as error bars. **A.** PPAR-α relative expression. **B.** PPAR-β relative expression. **C.** ACOX1 relative expression.

Two genes involved in xenobiotic biotransformations were tested. CYP1A1, involved in hydroxylation reactions of lipophilic substances in phase I, and GST, involved in conjugation of activated compounds to GSH in phase II (Aloke et al., 2024; McDonnell & Dang, 2013). No trends or significant responses were recorded following the GenX exposure (figure 14).

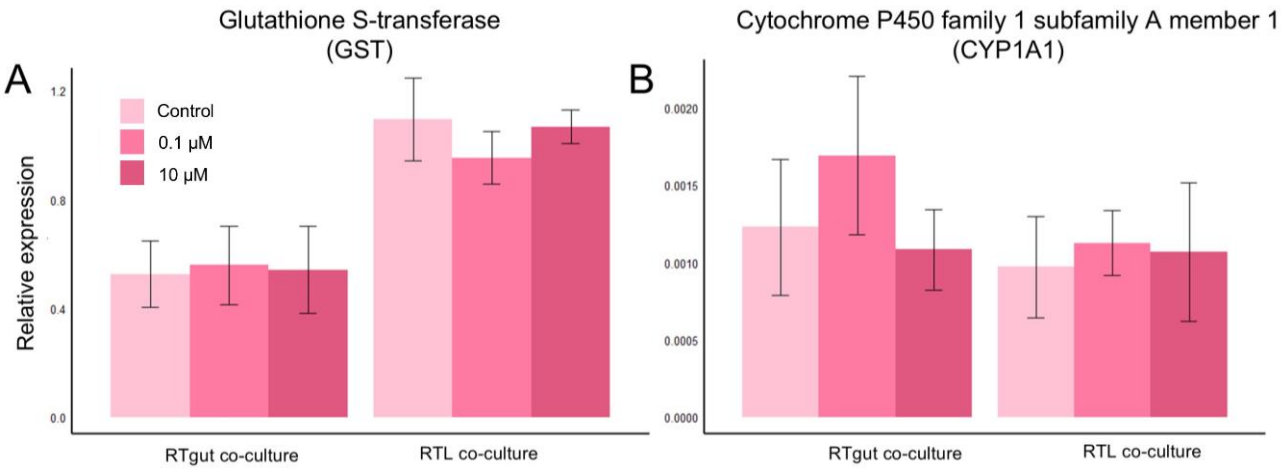

Figure 14. Relative expression of GST and CYP1A1 following GenX exposure. Bars represent the mean value of four or three replicate samples, SD shown as error bars. A. Relative gene expression for GST. No significant exposure related effects observed. B. Relative gene expression for CYP1A1. No significant exposure related effects observed.

No significant trends were detected for the co-cultured cells at either of the tested GenX concentrations. One additional co-culture exposure with a new stock of GenX was conducted at higher concentrations. Contrary to expectations cytotoxic screening of the new stock revealed no cytotoxicity in the previously tested concentration range. The exposed cells were harvested and lysates stored, but subsequent analyses were halted due to doubts about the integrity of the new stock. Focus was instead shifted to PFOA. The second GenX experiment was conducted in connection with, and according to the same protocol as the PFOA experiment described below.

3.2 PFOA trial

To increase the chances of gene expression responses, the highest exposure concentrations for PFOA was set to just below the EC10 concentrations. The co-culture setup was assembled, and exposure was done on day 22 with 10 µM and 100 µM PFOA, for 48 hours. The FBS-concentration in the basolateral compartment was kept at 5% throughout the whole establishment period and during exposure. In addition to the co-culture an RTL-W1 monoculture was exposed with the same concentrations, in L-15ex medium, for 48 hours.

The expression patterns of the three distinct cell groups: RTgutGC and RTL-W1 from the co-culture, and RTL-W1 from the monoculture, are presented side by side for each gene (figure 15, 16, and 17). Stars indicate the level of significance. Orange stars (*) indicate significance level based on the non-parametric Mann Whitney U tests. Black stars (*) indicate significance level of parametric tests: one factor ANOVA followed by Dunnett´s two-sided ad hoc test.

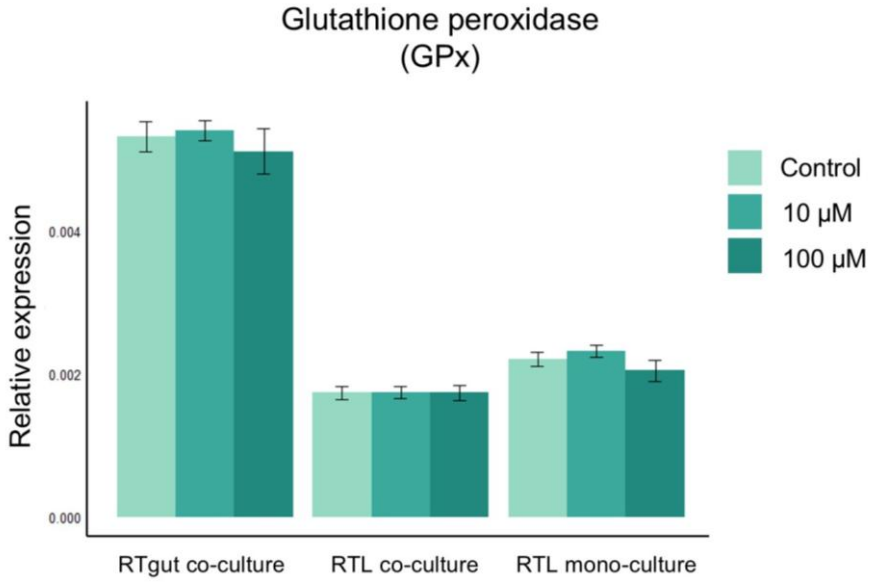

Figure 15. Relative expression of gluthathione peroxidase (GPx) on the co-culture set up and a standard RTL monoculture. Bars represent the mean value of 4 or 3 replicate samples, SD shown as error bars. The co-cultured RTgut showed the highest basal expression, as previously seen for the GenX trial in figure 12. No concentration dependent changes in expression levels could be observed.

For GPx, no exposure-related alterations in gene expression were observed; however, a significant difference in basal expression was evident between the gut and liver cell line (see figure 15). This observation aligns with the expression patterns observed in the GenX trial (see figure 12).

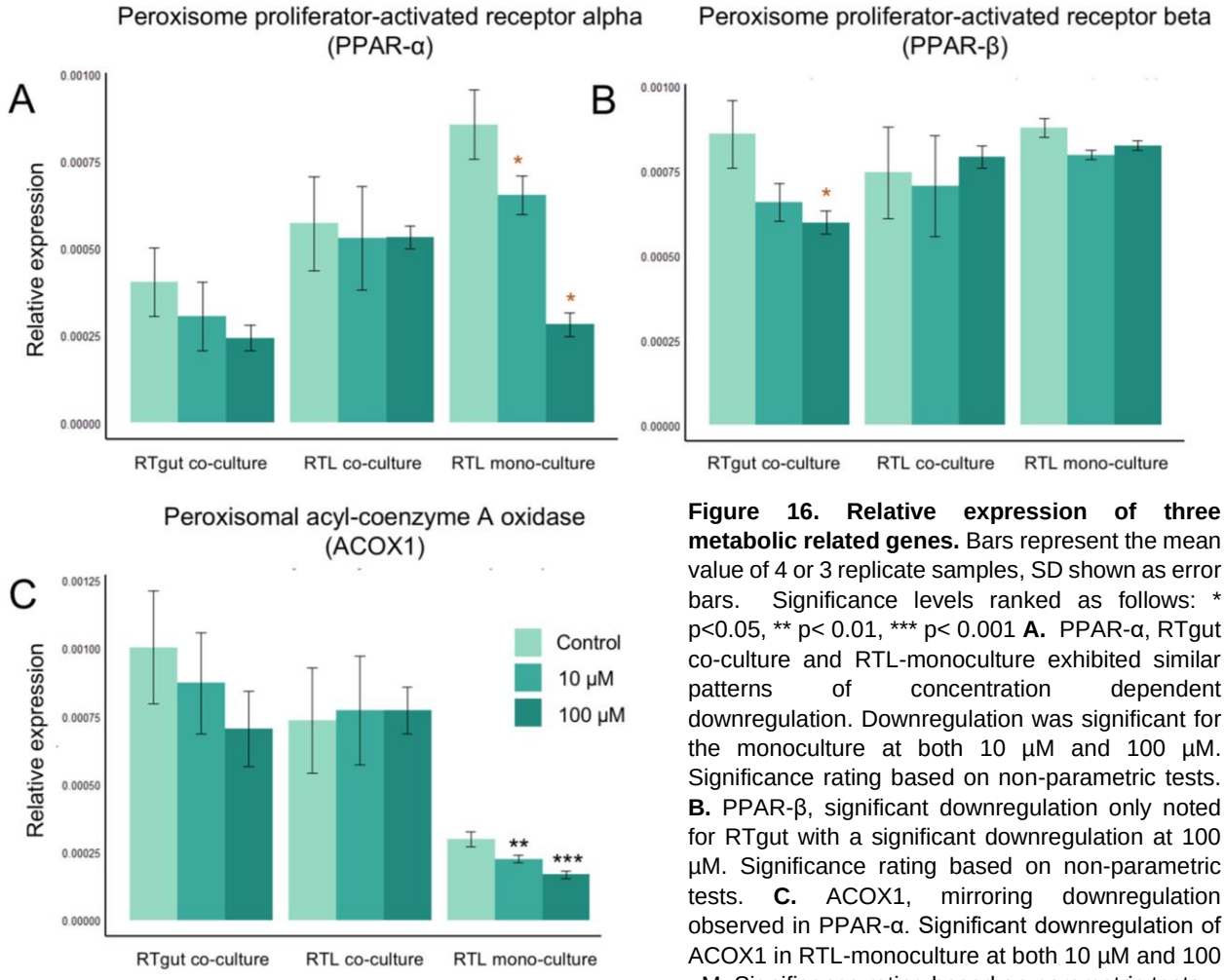

Figure 16. Relative expression of three metabolic related genes. Bars represent the mean value of 4 or 3 replicate samples, SD shown as error bars. Significance levels ranked as follows: * $p<0.05$, ** $p< 0.01$, *** $p< 0.001$ **A.** PPAR-α, RTgut co-culture and RTL-monoculture exhibited similar patterns of concentration dependent downregulation. Downregulation was significant for the monoculture at both 10 µM and 100 µM. Significance rating based on non-parametric tests. **B.** PPAR-β, significant downregulation only noted for RTgut with a significant downregulation at 100 µM. Significance rating based on non-parametric tests. **C.** ACOX1, mirroring downregulation observed in PPAR-α. Significant downregulation of ACOX1 in RTL-monoculture at both 10 µM and 100 µM. Significance rating based on parametric tests.

For the metabolic genes significant downregulations were observed. Distinct exposure related effects were observed for PPAR-α and ACOX1 in the monocultured RTL cells. A similar downregulation was noted for RTgutGC in the co-culture but could not be statistically supported due to the limited sample size and variability. The basal expression of PPAR-β was higher in co-cultured cells than for the α-isoform. Significant downregulation of PPAR-β was noted for RTgut co-culture (see figure 16).

For GST upregulation was observed solely in the monoculture at 100 µM. However, significant upregulation of CYP1A1 was found for both cell lines in the co-culture. A distinct difference in basal expression of the CYP gene was noted both between RTgut and RTL in the co-culture but also clearly between co-cultured and monocultured RTL cells. The substantially lower basal CYP1A1 for the monocultured RTL cells, compared to the co-cultured was observed for to two separate monocultures and two co-cultures, with RTL-W1 cells of different passage numbers.

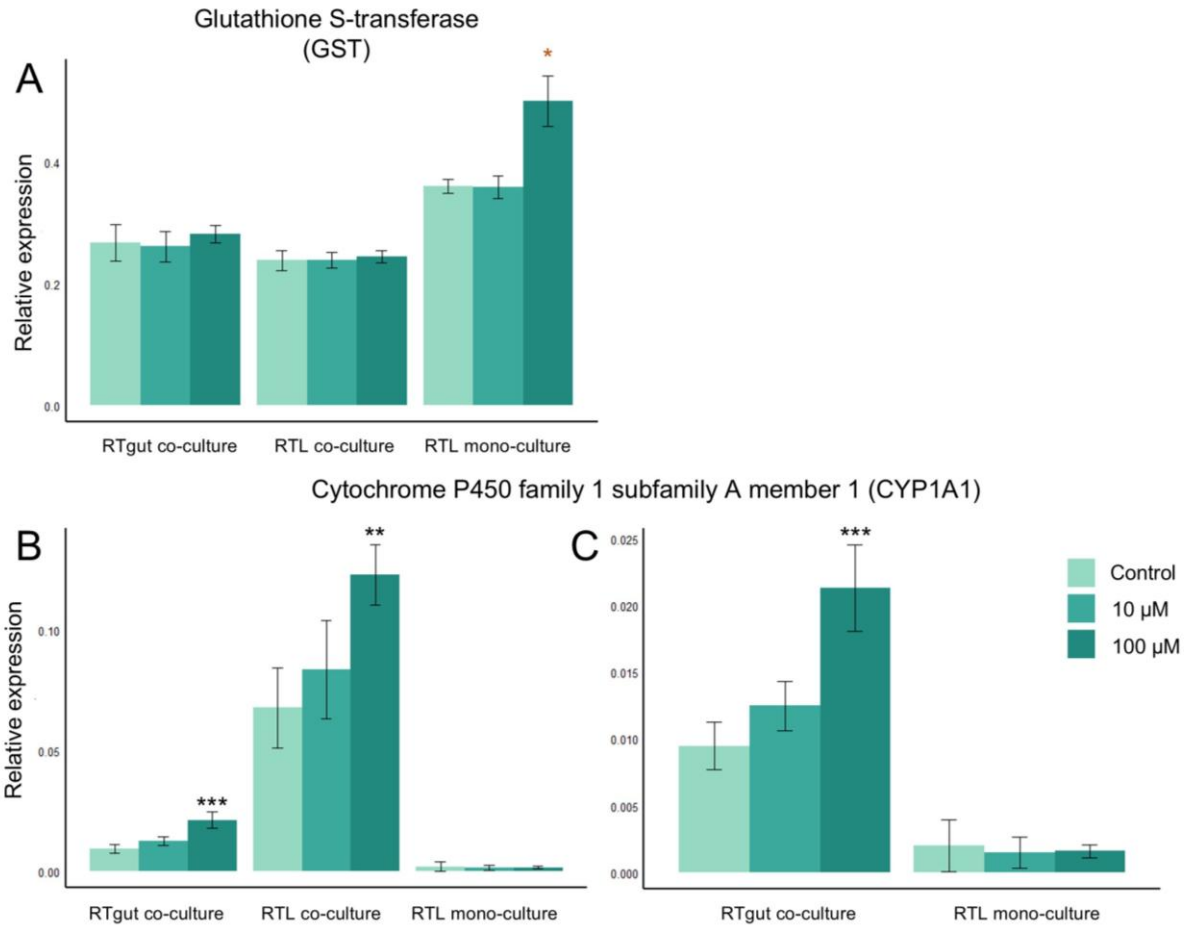

Figure 17. Relative expression of GST and CYP1A1. Bars represent the mean value of 4 or 3 replicate samples, SD shown as error bars. Significance levels ranked as follows: * p<0.05, ** p< 0.01, *** p< 0.001 **A.** Significant upregulation of GST for the monoculture at 100 µM. Significance rating based on non-parametric tests. **B.** Both co-culture and monoculture displayed in one plot. Similar upregulation of CYP1A1 in RTgut- coculture and RTL co-culture. Large differences in basal expression both between the cell types and two RTL setups. Significant upregulation at 100 µM for RTgut co-culture and RTL monoculture. Significance rating based on parametric tests. **C.** RTgut and RTL monoculture plotted together, without RTL co-culture for better visualisation. Low expression of CYP1A1 in the monocultured RTL and large variation across replicates.

4. PFOA feeding of rainbow trout

In the final part of the project, a feeding regime was developed for rainbow trout to attempt comparison between *in vivo* data and the results obtained *in vitro*. In addition to control cubes without PFOA, two exposure groups were created with different dosages: 1 mg PFOA or 10 mg PFOA/kg of fish body

weight. The exposure doses were chosen to be high enough to enable investigation of acute responses. Dosages were selected based on maximum concentrations measured in wild-caught fish (Barbo et al., 2023), as well as the PFOA dosage used by Arukwe & Mortensen (2011).

Ingestion of the food cubes was not forced and due to rejected food by some fish, the resulting sample size was smaller than initially planned. In total 14 fish were sampled (control: n=6; low dose: n=4; high dose: n=4). Tissue samples were homogenised; lysates were prepared and RNA extracted. qPCR analysis was conducted for the same genes analysed in the *in vitro* PFOA experiment. The relative gene expression for each gene presented below as bar plots (see figure 18,19, 20, 21, 22 and 23).

Elevated basal GPx expression in intestinal tissue compared to liver was observed. This finding aligns with observations *in vitro* with significantly higher basal expression for RTgutGC compared to RTL-W1 (see figure 12 and 15). High variability, especially in the control group, makes interpretations of trends precarious, however, results might suggest downregulation in liver with elevated PFOA dosage.

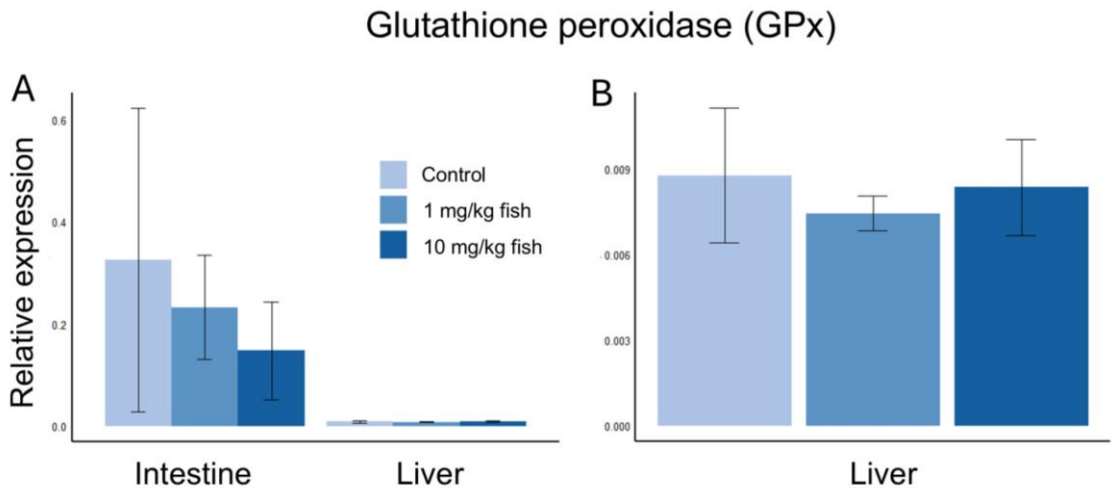

Figure 18. Relative expression of GPx in rainbow trout liver and intestinal tissue. Bars represent the mean of the biological replicates (n=4 for treatments an n=6 for control), SD shown as error bars. **A.** Marked difference in basal expression between intestinal and liver tissue. Great variance in the control for the intestinal tissue makes interpretations of trends challenging. **B.** Liver tissue GPx expression plotted by itself.

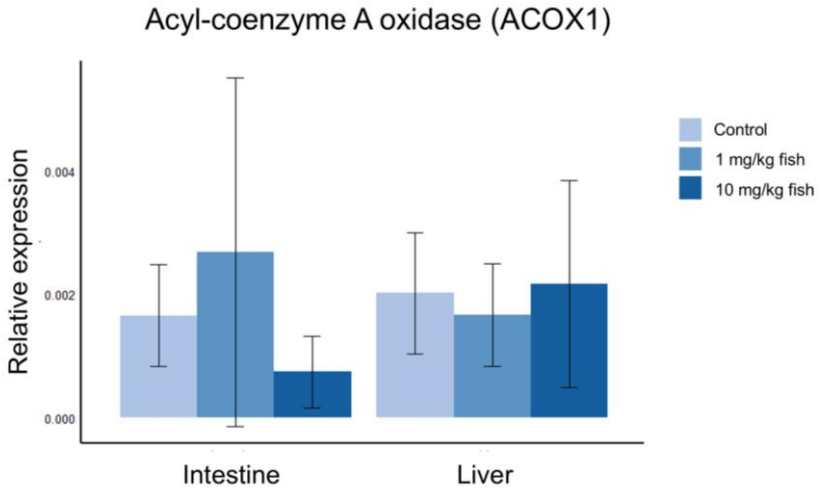

Figure 19. Relative expression of ACOX1 in rainbow trout liver and intestinal tissue. Bars represent the mean of the biological replicates (n=4 for treatments an n=6 for control), SD shown as error bars. Basal expression consistent across the two tissue types. Large variance within treatment groups, no significant treatment effect or distinguishable trends.

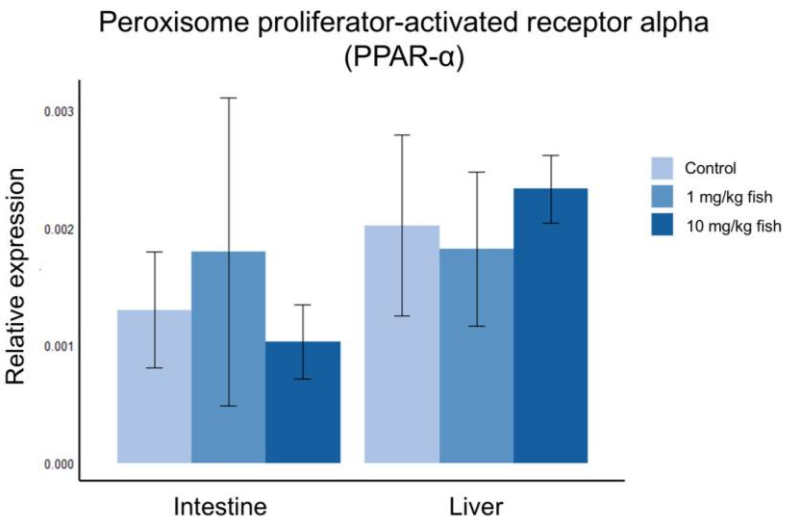

Figure 20. Relative expression of PPAR-α in rainbow trout liver and intestinal tissue. Bars represent the mean of the biological replicates (n=4 for treatments an n=6 for control), SD shown as error bars. Basal expression consistent across the two tissue types. No significant treatment effect or distinguishable trends.

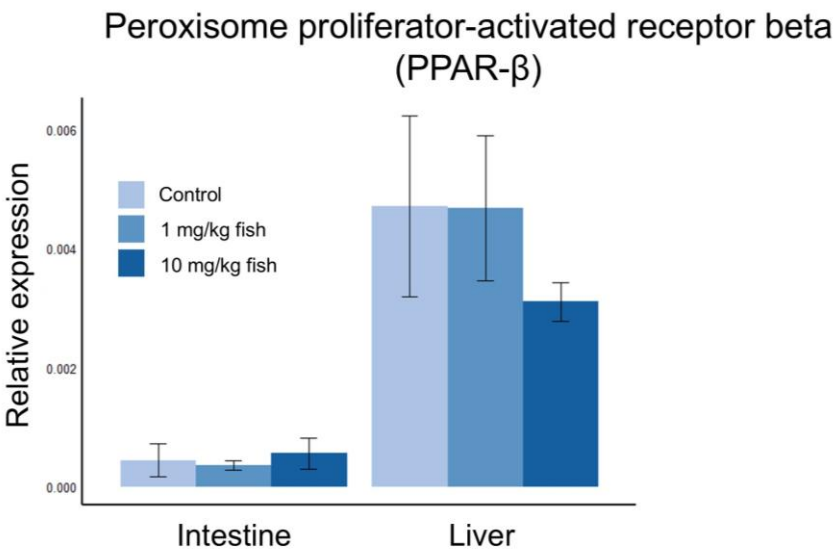

Figure 21. Relative expression of PPAR-β in rainbow trout liver and intestinal tissue. Bars represent the mean of the biological replicates (n=4 for treatments an n=6 for control), SD shown as error bars. PPAR-β basal expression significantly higher in the liver tissue compared to intestinal tissue. The results suggest a downregulation in the liver tissue at 10 mg/kg PFOA but could not be supported statistically. The low PPAR-β expression in the intestinal tissue renders the q-PCR data unreliable.

Contrary to the distinct downregulation observed *in vitro* no significant downregulation was noted for either of the genes related to lipid metabolism (figure 19, 20 and 21). A pronounced higher basal expression for PPAR-β was seen in liver compared to the intestinal tissue. Additionally, the expression levels measured in liver for PPAR-β suggest a downregulation of the peroxisome proliferation receptor gene at 10 mg/kg PFOA (figure 21).

For GST and CYP1A1, no significant changes in gene expression were noted 48 hours after the PFOA feeding. However, a slight, non-significant upregulation could be observed in the liver tissue for GST (see figure 22).

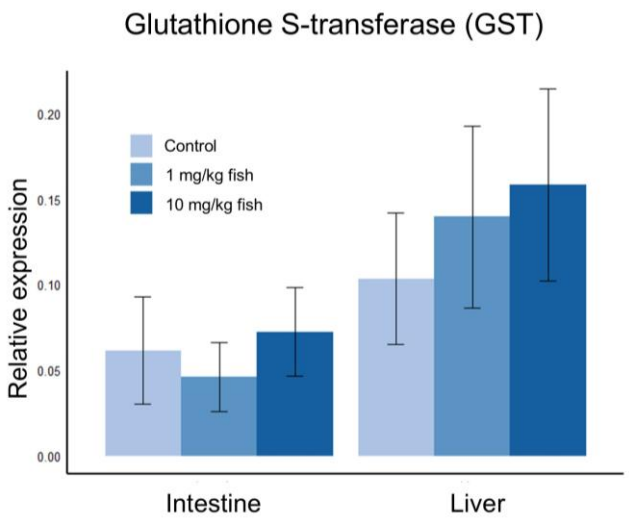

Figure 22. Relative expression of GST in rainbow trout liver and intestinal tissue. Bars represent the mean of the biological replicates (n=4 for treatments an n=6 for control), SD shown as error bars. The results suggest treatment related downregulation in the liver tissue but could not be supported statistically.

CYP1A1 expression levels differed significantly between intestinal and liver tissue. Expressions levels were markedly higher in the liver tissue (see figure 23).

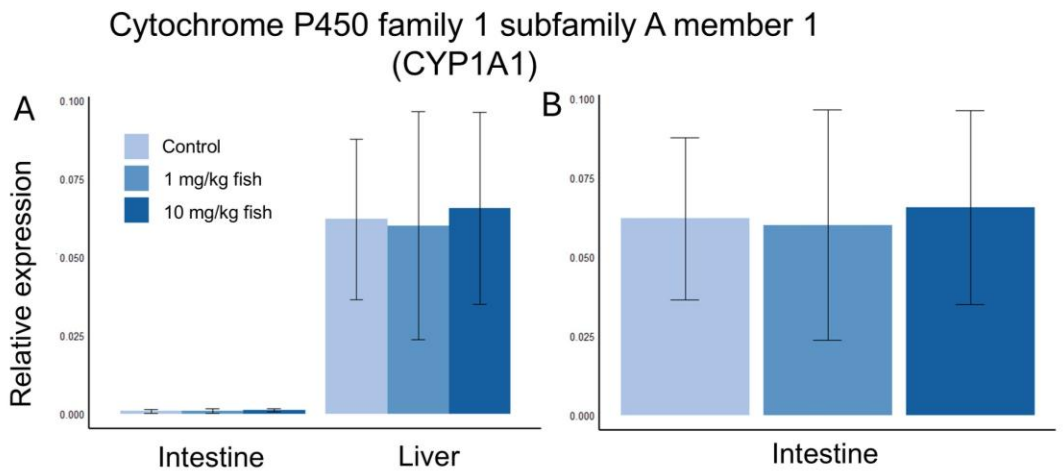

Figure 23. Relative expression of CYP1A1 in rainbow trout liver and intestinal tissue. Bars represent the mean of the biological replicates (n=4 for treatments an n=6 for control), SD shown as error bars. **A.** Basal expression of CYP1A1 significantly higher in the liver tissue compared to intestinal tissue. No distinguishable trends across treatment groups. Large variance noted for all treatment and tissue types. **B.** CYP1A1 relative expression for intestinal tissue, plotted alone for better visualisation.

5. CYP1A1 expression *in vitro- in vivo* comparison

Interest in CYP1A1 was prompted by to the marked difference in basal expression between RTL-W1 cells from the co-culture and cells from the monoculture.

To further investigates this finding an additional experiment was conducted where RTL-W1 was seeded using the same protocol as for the co-culture. 24 hours post seeding, either L-15ex or serum supplemented L-15 (5% FBS) was added, and the monocultures were incubated for 48 hours. Gene expression analysis revealed significant differences in CYP1A1 expression between the cells kept in the two different mediums (see appendix 4). However, even for the serum supplemented monoculture

CYP1A1 expression levels remained significantly lower than the co-cultured cells. Compared to the CYP1A1 levels observed *in vivo*, the co-cultured RTL-W1 exhibited similar expression levels whereas the monoculture expression levels were significantly lower. A bar plot comparing the *in vivo* expression with the different *in vitro* setups shown below in figure 24.

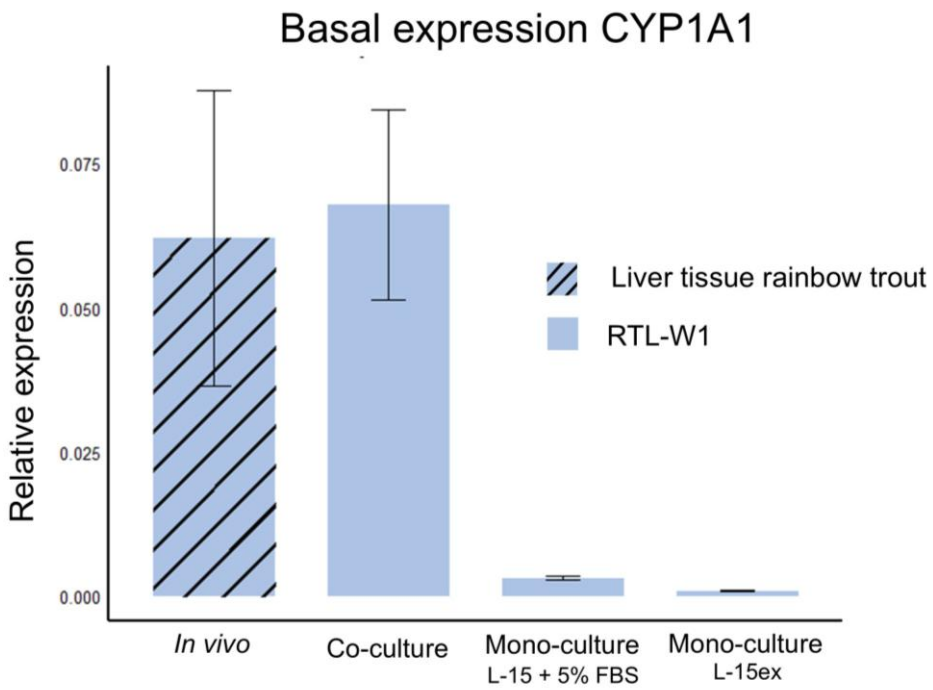

Figure 24. Comparison of basal expression of CYP1A1 between liver tissue from rainbow trout and three *in vitro* RTL-W1 setups, from left to right: liver tissue, RTL-W1 co-cultured together with RTgutGC in transwell inserts for 48h, monoculture 48h in FBS- supplemented L-15 medium, and monoculture 48h in L-15ex medium. Bars represent the mean of the biological/ technical replicates, SD shown as error bars. Significant difference in CYP1A1 expression between the cells from the monoculture compared to both monocultures. RTL-W1 expression levels comparable to *in vivo* levels. Comparable co-culture CYP1A1 expression levels have been observed for two separate co-culture trials.

Discussion

This project was conducted with the aim of evaluating a co-culturing system combining two rainbow trout cell lines for ecotoxicological research purposes. Assessments were mainly based on investigations of transcriptional responses to the perfluoroalkyl acid PFOA. In this section the results will be discussed and interpreted in the context of previously published data and existing literature.

TEER measurements confirmed a tight epithelial barrier for RTgutGC grown on porous supports. The epithelial resistance increased across 21 days reaching 97 ±18 Ω x cm^2 before exposure. This TEER progression exceeds previously reported values for similar RTgutGC models but remains within a physiologically relevant range for fish intestine.

Pioneering studies by Geppert et al. (2016), Minghetti et al. (2017) and Wang et al. (2019) introduced the RTgutGC epithelial barrier model and made progress in characterization of its polarization, permeability and ultrastructural features. These studies presented maximum TEER values between 30-50 Ω x cm^2. In live fish the intestinal epithelial barrier resistance has been reported to span a wider range, between 30-200 Ω x cm^2 (Claude & Goodenough, 1973; Loretz, 1995; Sundell et al., 2003). Thus, both the lower values recorded in earlier studies and the higher TEER recorded for the *in vitro* model developed in this study falls within a physiologically relevant range.

No unified protocol for a standardised RTgutGC barrier model has yet been established. Hence, no set benchmarks exist for the expected progression of epithelial resistance. Geppert et al. (2016) found inserts with 0.3 µm pore size to support the best barrier growth. On these inserts they saw a steady increase in epithelial resistance from 15 to 40 Ω x cm^2 across 35 days. Wang et al., (2019) observed similar TEER progression, reaching 30 Ω x cm^2 at day 28. The progressive TEER development aligns with transwell insert models developed for human intestinal cells (Darling et al., 2020; Hidalgo et al., 1989). Contrary to these findings Minghetti et al. (2017) observed no significant increase in epithelial resistance across 28 days but instead saw sustained TEER values around 35 Ω x cm^2 across 28 days. They did, however, observe a progression in polarised features that peaked at day 21.

According to guidelines developed by Claude & Goodenough (1973) the epithelial resistance range observed both *in vivo* and *in vitro* classify the fish intestinal barriers as “leaky”. In these types of membranes water transport is believed to be mainly paracellular, that is through the intracellular spaces between the epithelial cells. Sundell et al. (2003) characterized permeability in Atlantic salmon (*Salmon salar*) and recorded transepithelial resistance for the intestine. They revealed a shift in resistance along the digestive tract. For the posterior regions TEER around 30 Ω x cm^2 was recorded while 120-225 Ω x cm^2 was reported for posterior parts (Sundell et al., 2003). Based on this finding the lower TEER from previous studies represent a more permeable posterior intestine while the model developed in this study more closely resembles the anterior intestine.

It is possible that the higher TEER observed in this study stem from a difference in seeding density compared to earlier studies. Geppert et al., (2016) and Minghetti et al., (2017) seeded 6-well inserts at 62,500 cells/cm^2 while Wang et al., (2019) adjusted seeding density to 65,820 cells/cm^2 for the same insert size. The seeding density in this study was higher with 77,000 cells/cm^2 in 6-well inserts. Notably, Wang et al. (2019) reported double-layer formations for inserts grown beyond 28 days. It is possible that similar multilayering has occurred in this study, potentially at an earlier timepoints due to the higher cell density. Such multilayering of cells could explain a higher epithelial resistance and the leap in TEER observed between week 2 and 3 (see figure 6).

While TEER is a well-established and sensitive method for measuring epithelial resistance there are inherent limitations to the electrode setup (Srinivasan et al., 2015). The so-called chopstick electrode utilised in this study is a handheld device relying on the user to place and hold the electrodes in the two compartments while simultaneously monitoring the oscillating measurements on the display. Despite careful use, the electrode handling itself likely contributed to the variance in TEER recorded in this study.

Further, the small electrodes have a limited area in which they can deliver a uniform current. According to Srinivasan et al. (2019) the electrodes will overestimate the TEER in inserts with a culture area bigger

than 24 mm^2. Instead, an Ohm-chamber is proposed for bigger inserts, such as the ones used in this study. In the Ohm-chamber dish electrodes on either side of the membrane measure the resistance. This setup was used by Geppert et al. (2016) and it successfully recorded TEER measurements with low variability.

Despite the limitations of the chopstick electrodes, they provided solid indications that the membrane model was successfully established, with TEER values in a physiologically relevant range. However, to increase throughput, enable experiments of larger scale, and improve the sensitivity of TEER measurements, alternative electrode types should be considered.

Further characterization possibilities for the gastrointestinal barrier model.
Further characterization of the barrier was beyond the scope of this study but is advisable. Given the model's novelty, the lack of standardization, and the limited precision in the TEER measurements, complementary tests could strengthen the characterization of the established barriers. Options for additional, non-invasive methods are limited but multiple analysis methods could be run alongside the main experiment. One example of such approaches is quantification of the expression of tight junction genes (for example ZO-1) for RTgutGC grown in the transwell inserts compared to RTgutGC from simple monocultures (Minghetti et al., 2017). Elevated expression levels of the tight junction gene could indicate an enhanced barrier function. Additionally, electron microscopy could be used to investigate the membrane morphology and the previously discussed possibility of a multilayer formation.

Polarisation of the membrane could also be investigated via antibody staining followed by fluorescent microscopy or scanning and transmission electron microscopy (Minghetti et al., 2017). Apical localization of tight junction proteins and basolateral distribution of Na^+/K^+-ATPases are known features of a polarised membrane (Geppert et al., 2016; Minghetti et al., 2017). Permeability could be analysed via fluorescently labelled dyes added directly into the insert compartment. The medium of the receiving compartment below the inserts could be sampled at various timepoints and the resulting fluorescence give indications of the membranes permeability (Minghetti et al., 2017). Similar permeability studies can be performed with radioisotopes for improved accuracy (Srinivasan et al., 2015).

The proposed additional characterisation efforts would not only strengthen the evidence supporting the claim that the model was successfully established, but also lay the foundation for future toxicokinetic and mechanistic studies of the full co-culture setup.

The cytotoxicity screening established a relative toxicity ranking of the PFAAs.
PFBS, the short chained PFSA showed low cytotoxicity in the RTgutGC whereas PFOA and GenX exhibited measurable toxicity within the tested range. The relative toxicity of the tested PFAAs was GenX > PFOA >> PFBS (figure 8). This indicates a higher cytotoxicity for GenX, the substitution chemical, compared to the old-school PFOA. It should be noted, however, that the dose-response data from the initial screening were not of optimal quality. Furthermore, GenX and PFBS screenings has not yet been repeated, limiting the reliability of the results.

Ranking of the relative potency of PFAAs becomes complex when different functional groups, accumulation potentials and elimination rates are considered. The relative cytotoxicity ranking of GenX > PFOA observed in this study does, however, correspond to findings by Gomis et al. (2018). Gomis et al. (2018) utilised publicly available data for PFAA exposure in rats to evaluate the toxic potency of long chained PFAAs and substitution fluoroalkyls. When they corrected for toxicokinetic factors it was revealed that the toxic potency of the substitution chemicals did not differ significantly or, in the case of GenX, even exceeded that of their predecessor compounds (Gomis et al., 2018). The same trend was observed by Solan et al. (2022) for the HepaRG human liver cell line with GenX being more cytotoxic than PFOA. However, the reverse relationship was found for other human cell lines (Solan, Senthilkumar, et al., 2022).

The relative potency of GenX and PFAO is not firmly established in the literature and appears to differ between species, cell lines and experimental designs. Nevertheless, the apparent EC50 of PFOA and

GenX being in the same order of magnitude is supported by previous attempts to rank their potency (Gomis et al., 2018; Satbhai et al., 2022a; Solan, Senthilkumar, et al., 2022).

EC50 values for PFOA were established for RTgutGC and RTL-W1.
For the refined cytotoxicity screen, a Weibull regression model was fitted to the dose-response curves to determine EC50 values. Sensitivity varied slightly between the two rainbow trout cell lines with EC50 values at 295 µM for RTgutGC and 328 µM for RTL-W1. This suggests higher sensitivity for RTgutGC. However, based on the EC10, 97 µM for RTL- W1 and 197 µM for RTgutGC it could be argued that RTL-W1 is more sensitive as viability loss is recorded earlier for the liver cells compared to the gut cells.

Currently, no cytotoxicity data is available for PFOA exposure on the specific rainbow trout cell lines tested in this study. However, the EC50 values can be contextualized by data from other biological models. Zebra fish (*Danio rerio)* embryo/larval tests report embryotoxicity of PFOA within a similar cytotoxicity range as the rainbow trout cell lines. Gabreab et al. (2020) reported the median lethal concentration (LC50) of PFOA to be 232 µM for embryos 24 hours post fertilisation, while Satbhai et al. (2022) reported LC50 226 µM based on predicted internal concentrations for the same endpoint (Satbhai et al., 2022b).

Solan et al. (2022) reported an EC50 of 21 µM after 24 hours of PFOA exposure to mammalian gut cells, approximately one order of magnitude lower than for RTgutGC. The reduced sensitivity of the piscine cell lines, compared to mammalian cells aligns with the observation that piscine cell lines generally exhibit lower sensitivity compared to mammalian counterparts. This discrepancy has been attributed to slower metabolism and cell cycle progression in the fish cells (Castaño & Gómez-Lechón, 2005).

The applicability and predictive strength of *in vitro* models in ecotoxicology mainly depends on how well they mirror *in vivo* responses. A key limiting factor for cell-based *in vitro* models is their reduced sensitivity compared to whole organisms. Tanneberger et al. (2013) evaluated the predictability of the cytotoxicity assay with AlamarBlue/CFDA-AM on RTgill-W1. Testing a wide variety of compounds, they found *in vitro* EC50 values to generally be within a 5-fold range higher than corresponding *in vivo* data. Outliers, falling far outside the predicted discrepancy range were compounds of high hydrophobicity, volatility and toxicants that are metabolically activated (Tanneberger et al., 2013). No PFAS were tested but due to their amphiphilic nature and non-specific modes off action PFAS likely challenge a simple, universally applicable five-fold *in vitro* to *in vivo* conversion. Therefore, further comparative studies focusing specifically on PFAS are needed to better assess *in vitro* fish cell line predictability.

No significant treatment related effects were observed for the GenX exposed co-culture.
The absence of gene expression responses might suggest that the dose response curve from the initial screening overestimated the potency of GenX. Consequently, the chosen exposure concentrations may have been too low to trigger sublethal effects. Kinetic factors like the fluorochemical´s absorption and extraction rate may also have affected the gene expression results. Given GenX´s lower bioaccumulation potential and more rapid excretion rate compared to longer chained PFAAs, it is possible that the window for detecting sublethal effects is relatively narrow (Satbhai et al., 2022a). Nonetheless, the GenX trial did help in developing a logistically feasible workflow and confirmed sufficient RNA-concentrations and high sample quality. However, the absence of clear treatment related effects limits its usefulness for further evaluation of the co-culture method.

As the laboratory work progressed, the focus shifted to perfluorooctanoic acid (PFOA), the long-chain PFCA. PFOA already has a well-established toxicity profile which makes it a suitable model compound for method development purposes. Direct comparison between the first co-culture and the later co-culture setups is precarious as differences in culture media and time points for exposure introduce confounding factors. Additionally, the first co-culture exhibited, compared to the subsequent trials, lower TEER values throughout the establishment phase and greater variance overall.

Downregulation of the lipid metabolism-related genes PPAR-α and ACOX1 suggests lower responsiveness of the co-culture system compared to the monoculture.

The expression pattern observed across the three cell groups provide insight into how the PFOA exposure effected the cells differently depending on their exposure situation. The downregulated trend was evident both for RTL-W1 in the monoculture and for RTgutGC in the co-culture (see figure 16). However, the trend was statistically significant only for the monoculture which indicates a difference in responsiveness between the two culturing setups. The discrepancy may also reflect inherent sensitivity differences between RTgutGC and RTL-W1, as suggested by the cytotoxicity screening (see figure 11). Basal expression levels further indicates that the monoculture has a higher baseline activity of PPAR-α, possibly due to the minimal medium.

In addition, the formation of the RTgutGC barrier has previously been shown to reduce the sensitivity towards toxicants, compared to simple monocultures. This may be attributed to the higher cell density in the barrier which limits exposure area, and increased efflux activity in the polarised membrane (Minghetti et al., 2017). Ideally, reduced responsiveness of RTgutGC in the inserts should be confirmed by direct comparison with an RTgutGC monoculture.

The absence of PFOA related effects for the RTL-W1 in the co-culture is likely attributed to its location in the receiving, basolateral compartment. These RTL-W1 cells are solely receiving exposure medium that has first crossed the gut cell barrier. The complete absence of gene expression alterations thus suggests low permeability of the gut cell barrier for PFOA. Moreover, the medium in the basolateral compartment is supplemented with serum, which is believed to reduce cell sensitivity and decrease the availability of PFOA, which will be discussed later.

Investigations of the permeability of the gut cell barrier and chemical analysis of the medium in the donating vs the receiving compartments could further strengthen interpretations of the distribution of PFAA in the co-culture. Additionally, obtaining permeability data on different types of toxicants of varying polarity and size would contribute with valuable information on the membranes transporting abilities and available transport routes through the barrier.

The low permeability and shielding effect caused by the gut cell barrier could limit the co-culture systems usefulness for short, acute exposure tests where responses of the liver cells are of interest. However, longer sustained exposure could be an interesting application for the stable co-culture system. This would allow for studies of toxicokinetic outcomes in more chronic exposure situations. Generally, the long establishment period and the suggested lower sensitivity compared to monocultures makes the co-culture less suited for high-throughput toxicant screenings and more appropriate for targeted toxicokinetic or mechanistic studies.

Still, the systems sensitivity must be evaluated at environmentally relevant concentrations. The concentrations used in this study are manyfold higher than concentrations recorded in biota and other environmental compartments (Barbo et al., 2023; Dagorn et al., 2024). The unrealistically high exposure concentrations were chosen for the purpose of obtaining clear results to support the early phases of method development. However, to further advance the evaluation of the model's suitability for ecotoxicological studies the sensitivity to more environmentally realistic exposure scenarios need to be established.

The metabolic downregulation partly contrast finding reported in previous studies.

A marked reduction in PPAR-α expression was mirrored by a similar downregulation in ACOX1 following PFOA exposure. The two genes parallel downregulation supports their established interconnection (Arukwe & Mortensen, 2011; Vluggens et al., 2010b). Previous studies suggest that PFOA can act as a PPAR-α substrate, prompting binding to response elements in the DNA which in turn upregulates target genes such as ACOX1. The connected expression patterns observed for both the receptor and the target gene thereby aligns with their regulatory relationship. However, the concentration-dependent downregulation induced by PFOA diverge from the sublethal effects often described where upregulation,

rather than downregulation of lipid metabolism is reported (Arukwe & Mortensen, 2011; J. W. Lee et al., 2020).

Cellular responses to PFAS are known to be multifarious and expression patterns for individual or pairs of genes could be ambiguous. Fluorochemicals are thought to influence multiple cellular systems, among other the mitochondrial membrane function, the thyroid axis and immune related enzymatic systems, all of which are interconnected with the metabolic regulation to varying degrees (Ma et al., 2024). It is therefore challenging to reach a conclusion on the reasons behind the downregulation without additional studies of the interrelated systems. The complex, interconnected relationship between the PPARs and other cellular pathways highlights the lipid metabolic genes´ shortcomings as clear, distinctive biomarkers.

Important to note is that the presented results solely offer a snapshot of the cellular RNA transcription, 48-hours post exposure. The exposure time was selected arbitrary but with the intention to allow for cell-cell interactions in the co-culture and with regards to the piscine cell lines relatively slow metabolism. It is plausible that an initial PPAR-α activation did occur within the first hours of exposure but was not captured. At 48 hours feedback loops could have shifted the expression patterns. A temporal analysis, studying gene expression across multiple time points would greatly improve the possibility to interpret the cellular mechanisms at play. It would not only help with interpretations of the effects of the PFAA exposure but also of interactions between the two co-cultured cell lines.

Results do not imply induction of oxidative stress.
Induction of the antioxidant enzyme glutathione peroxidase (GPx) is a classical cellular response to oxidative stress. However, no upregulation of GPx was observed for either co-culture or the monoculture (see figure 15). The absence of an oxidative stress response aligns with the observed downregulation of lipid metabolism. A suppressed ACOX1 expression suggests a reduced electrons flow to the electron transport chain, and in turn lower reactive oxygen species (ROS) production.

GPx is a link between the targeted enzymatic antioxidant defence and the broader non-enzymatic glutathione (GSH) based antioxidant system. GSH plays a dual role, both of which are potentially related to cellular PFAAs responses. A decrease in GSH levels may therefore indicate either ongoing oxidative stress response and/or activation of the xenobiotic metabolism. In the case of oxidative stress GSH is oxidized by ROS to its GSSG form in a redox reaction mediated by GPx. GSH is also required for certain conjugation reaction in phase II of biotransformation, facilitated by GST. Notably, GST was significantly upregulated in the monoculture at 100 µM of PFOA (see figure 17). The lack of GPx response, together with the GST induction therefore suggests activation of xenobiotic metabolism rather than oxidative stress responses.

It is important to keep in minds that the qPCR data reports on the cellular mRNA levels, not the subsequent quantity of translated proteins or their activity levels. Consequently, the presented results indicate the transcription state just before the cell harvest and not the actual intracellular levels of antioxidants. Characterisation of the antioxidant levels and the GSH/GSSG ratio would therefore provide valuable additional insights, enabling further characterisation of potential oxidative stress responses.

Significant upregulation of CYP1A1 was observed in the co-culture.
CYP1A1 encodes a cytochrome P450 enzyme from family 1, subfamily A. Enzymes from the CYP1 family are inducible and take part in phase I biotransformation reactions where they activate substrates in preparation for further detoxification transformations. CYP1A expression is initiated by ligands binding to the aryl hydrocarbon receptor (AhR) (Iacopetta et al., 2023; McDonell & Dang, 2013).

CYP1A is traditionally considered a biomarker for aromatic hydrocarbons and the pronounced CYP1A1 activity in response to PFOA is therefore somewhat unexpected. In addition to not being a traditional inducer of the AhR-receptor the highly stable PFOA molecule is resistant to biodegradation. No biotransformation by the CYP enzymes is therefore expected for PFOA (Kolanczyk et al., 2023). Despite this, AhR activation has been reported for fish in response to PFAAs in several studies (J. W. Lee et al.,

2020; Y. Liu et al., 2008; Mortensen et al., 2011). This may be explained by the receptor´s low substrate specificity and the relative structural similarity between aromatic hydrocarbon and the long chained fluoroalkyls (Mortensen et al., 2011).

Interestingly, upregulation of CYP1A caused by PFOA has been documented in salmonid fish. Mortensen et al. (2011) performed a PFOA dietary exposure experiment on Atlantic salmon and reported increased transcriptional levels of CYP1A1 following PFOA exposure. The fish were dosed every third day. Upregulation was noted after the first and second dosage but was then reported to decrease below control levels. The time- and dose-dependent response found by Mortensen et al. (2011) could explain contradictory CYP1A1 expression patterns described for PFOA in literature (J. W. Lee et al., 2020).

CYP1A1 expression patterns were found to be markedly different between co-cultured and monocultured cells.

In the monocultured RTL-W1 cells, no changes in transcriptional levels were observed at either of the PFOA concentrations. In contrast, the co-culture system showed significant upregulation of CYP1A1 for both RTgutGC and RTL-W1. Notably, this was the only treatment related effect recorded for the co-cultured RTL-W1. The basal expression of CYP1A1 in the co-cultured RTL-W1 cells was elevated compared to RTgutGC, and interestingly also compared to the monocultured RTL-W1 (see figure 23). The significantly higher overall CYP1A1 transcriptional activity raised questions regarding the two culturing environments influence on the xenobiotic detoxification response levels.

Both RTL-W1 populations were from the same passage number, seeded identically in the 6-well plates and grown for the same duration before harvest. However, a key confounding factor in the comparison was the difference in medium. The exposure medium was applied directly to RTL-W1 in the monoculture while the RTL-W1 in the coculture was only indirectly exposed via the gut cell barrier. Instead, the co-cultured RTL-W1 cells were in serum-supplemented medium.

PFOA is bioaccumulative and known to accumulate particularly well to serum proteins (Tanneberger et al., 2013). To minimise medium-induced effects on the PFOA availability exposure solutions were therefore prepared in minimal serum free medium. Additionally, serum rich medium is known to protect the cells, making them less responsive. L-15ex on the other hand can induce a more energy-conserving state in the cells, which makes them less resilient when exposed to xenobiotics (Tanneberger et al., 2013). The low inherent sensitivity of piscine cell lines thereby prompted the use of minimal medium for the exposure solutions. Additionally, during the cytotoxicity screenings the cells were exposed to PFAAs in L-15ex, as suggested by guidelines from OECD test no. 249. Therefore, using the same medium for the co-culture and monoculture exposures allowed for more accurate predictions of cell viability.

An additional monoculture experiment was conducted to investigate medium induced effects on the CYP1A1 expression. This trial showed a significant difference in basal expression between RTL-W1 cultured 48 hours in FBS-supplemented medium versus RTL-W1 cultured in L-15ex. This suggest that the serum supports a more active xenobiotic detoxification state. However, the medium-induced difference was marginal compared to the pronounced difference observed between co-cultured and monocultured cells, regardless of medium composition (see figure 24).

Elevated CYP1A1 levels could indicate enhanced activity of xenobiotic systems in the co-cultured RTL-W1.

Similar marked difference in CYP1A1 transcriptional levels in co-cultured RTL-W1, compared to monocultures was documented by a previous student in the lab, 10 years ago (T. Catinot, (2014), unpublished results). Upregulation of CYP1A1 transcriptional activity was also found by Krøvel et al. (2011) in a co-culture of primary hepatocytes and monocytes from Atlantic salmon. Moreover, increased basal expression of CYP1A1 has been reported by Lammel et al. (2019) for their novel RTL-W1 spheroid model. The cells in this 3D culture regained some ultrastructural features like tight junctions and desmosomes, making the cells more closely resemble the original hepatocyte phenotype. Lammel et al. (2019) suggest that the CYP upregulation points to an increased xenobiotic competence for the 3D-cultured induced by the improved, more *in vivo*-like microenvironment.

Building on the theory proposed by Lammel et al. (2019), the elevated transcriptional CYP1A1 level observed in this study suggests that the presence of the gastrointestinal RTgutGC barrier model altered the microenvironment for the liver cells below. The altered microenvironment in turn could have prompted the increased CYP-activity in the RTL-W1. One possibility is that cell-cell interactions between the two co-cultured cell populations triggered this change. For instance, excreted signalling molecules or waste products from the RTgutGC gastrointestinal barrier could have induced the de-differentiation of RTL-W1.

Characterization of the excretion products could support further investigations into the mechanisms behind the document heightened CYP-expression. Extraction of the exosome-fraction from the basolateral medium followed by characterization of its content is one possible route. Metabolomic studies is another possibility to map out all metabolites present in the medium.

***In vitro* to *in vivo* comparisons were limited.**

Unlike the previously mentioned studies that solely compared gene-expression between mono- and co-cultures, we were able to do additional comparisons to *in-vivo* transcriptional levels. In figure 22 the basal expression of CYP1A1 is compared between liver cells from rainbow trout liver tissue and the RTL-W1 cell line. The results suggest a similar expression level between the co-cultured cells and the liver tissue. This may strengthen the theory that increased xenobiotic competence could be connected to regained, more *in vivo*- like phenotypic features in the co-cultured cell lines.

Further quantitative comparisons between PFOA induced response *in vivo* and *in vitro* was not possible, mainly due to the difficulty of relating *in vitro* exposure concentrations to *in vivo* feeding dosages. Correlating concentrations between direct exposure of cultured cells and dosages feed to live animals is a difficult feat. The differences in level of biological organisation, exposure route, bioavailability, toxicokinetic factors and biological time scales are just some of the factors at play. However, a potential approach could be to measure intracellular concentrations in the cells and serum levels in the fish and thereafter compare the exposure levels. That was beyond the complexity of this study but would be a possible approach to enable quantitative evaluation of the co-culture versus *in vivo* responses in the future.

The approach adopted in this study was instead to aim for a high PFOA exposure in both cells and fish and then compare trends in a qualitative manner. However, no significant treatment related trends were observed for the *in vivo* data, mainly due to the high variance and small sample size. Despite this some analogue patterns were seen both *in vitro* and *in vivo*. An upregulation of GST in response to PFOA was observed in the liver tissue (see figure 22), which aligns with the heightened activity of xenobiotic biotransformation activity observed *in vitro* (figure 17). In contrast to the *in vitro* data no upregulation of CYP1A1 was observed *in vivo* (figure 23). Indications of ACOX1-downregulation in the intestinal tissue were seen when comparing the highest dose to the control (figure 19). This supports the observed *in vitro* downregulation of lipid metabolism.

Unfortunately, the reliability of any analysis of the *in vivo* trends is low. The variance is high, especially in the intestinal tissue data. This may partly be related to the chosen sampling technique. Scrapings of the intestine was performed, and only a small part of the scarping was then used for RNA-extraction. High heterogeneity of cell types in the samples is therefore likely.

The low sample size can be attributed to the design and outcome of the feeding regime. The feeding was not based on oral gavage (force feeding) but instead on voluntary ingestion of the prepared food. Oral gavage was avoided due to ethical reasons, to spare the fish from additional stress and save on labour hours. Unfortunately, the fish did not consistently accept the prepared gelatinous food cubes which resulted in the smaller than planned sample size. The refusal could be due to the cubes relatively large size, its consistency or the taste. These limitations should be considered in the planning of future feeding regimes, especially if multiple feedings in a timed series are required.

Conclusions & Outlook

In this study, a co-culture model combining RTgutGC and RTL-W1 cells was evaluated. More precisely the suitability for studies of sublethal, transcriptional-level changes was assessed via exposure of the model to the perfluoroalkyl acid PFOA. The transcriptional responses were compared to both monocultured cells and *in vivo* data. The results revealed limiting factors as well as promising potential for the co-culture model. While the project revolved around PFAS induced transcriptional responses, the primary focus of this study has not been on the cellular mechanisms behind the fluorochemical toxicity but rather on utilizing them as model compounds supporting the evaluation of the piscine cell line methodologies.

Firstly, the RTgutGC gastrointestinal barrier was successfully established and the model showed potential for supporting future *in vitro* studies of toxicokinetic processes in the fish intestine. However, the need for further characterization of the intestinal barrier became evident when attempts were made to compare epithelial resistance data to the limited available literature on similar cellular systems.

Secondly, transcriptional level effects were investigated. Significant translational responses were found in both the co-culture and monocultures for PFOA. These effects allowed for comparisons between the vulnerability of the different cell culturing environments, both between the two compartments of the co-culture and between mono- and co-cultures. Notably, the RTgutGC barrier showed low permeability for PFOA with subsequent low toxicant response for the RTL-W1 in the basolateral compartment. While the system may allow for studies of extended exposure situations the model's utility for short-term exposures could be limited by the low responsiveness. Further refinement and validation are needed to evaluate if the system is sensitive enough to detect transcription level effects at environmentally relevant concentrations of PFAS.

In addition, a marked increase in CYP1A1 transcription was observed in co-cultured RTL-W1 compared to monocultures. This observation aligns with similar observations in other co-culture or 3D culture systems and may indicate an enhanced xenobiotic metabolic competence for the cell lines in the co-culture model. Remarkably, basal CYP1A1 transcriptional levels in RTL-W1 from the co-culture resemble the *in vivo* levels observed in rainbow trout liver tissue. This notion shows great promise for the model's applicability as an improved *in vitro* platform for studying biotransformation-related toxicological processes. Further studies of the co-culture model built on this finding is thus of interest.

However, comparison of the *in vitro* transcriptional trends to data from the *in vivo* PFOA feeding experiment was proven to be difficult. The small samples size and high variability made it difficult to detect clear treatment related effects. While it limits the reliability of the in *vitro* to *in vivo* comparison it highlights the inherent challenges of *in vivo* studies. Further, this underscores the importance of continued advancement of *in vitro* alternatives.

In conclusion, this thesis work demonstrates that piscine *in vitro* models hold promise for ecotoxicological studies less reliant on animal testing. However, no single *in vitro* system will be universally applicable, allowing accurate screenings of all environmental contaminant classes and all biological systems. Future development efforts of the co-culture model should therefore not strive to achieve the broadest utility but rather focus on defining limitations and identifying appropriate applicability domains. The real power of *in vitro* methods comes from strategic applications, enabling the utilization of each model's strengths. This novel co-culture system should therefore be treated as one tool that, in combination with other methods, enables robust assessments of ecotoxicological effects.

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Appendix

Appendix 1. qPCR primers

Gene name	Sequence			Codes for
	Forward	Reverse	Reference	
GPx	´5´- CCT GGG AAA TGG CAT CAA GT-3	´5´- GGG ATC ATC CAT TGG TCC ATA T-3	Gunnarsson et al., 2009	Glutathione peroxidase, an antioxidant involved in the detoxification of hydrogen peroxide.
GST	5´- ACC TGG TGC TCT GCI CCA GTT-3´	5´- AGA GCT CAG GAA GCC CTT GAT-3´	Gunnarsson et al., 2009	Glutathione-S-transferase
SOD Cu	5´- CCA ATG AGC TTC ACA GGA CCA T -3´	5´- AGG CTG TTT GCG TGC TCA A-3´	Gunnarsson et al., 2009	Superoxide dismutase
SOD Mn	5´- CCA CAC CAT CTT CTG GAC AAA C-3´	5´- GGA GCC AAA GTC ACG GTT GA-3´	Gunnarsson et al., 2009	Superoxide dismutase
CYP1A1	5´- TCC TGC TCA CCA TCC CAG ACT GCA C - 3´	5´AGG ATG GCC AAG AAG AGG TAG ACC TC- 3´	Gräns et al., 2010	Cytochrome P4501A (CYP1A).
PPARα	5´- AAC GGC TCT GAA GGA TCC AC- 3	5´- CCG AAA GAA ACC CTT GCA GC-3´	Made with NCBI "Pick prime" for NM_001197211	Peroxisome proliferator-activated receptor alpha
PPARβ	5´-GCT GAT CAA TGG CAC GAC CC-3´	5´-CTG GTC GTT CAG GAA GAG GTC -3	Made with NCBI "Pick prime" for NM_001197207	Peroxisome proliferator-activated receptor beta
ACOX1	5´-GGC ATC GAG AAG ACC AAA AGA AG-3´	5´- CAG GAA CAT TGC CAG GTG GAG A-3´	Made with NCBI "Pick prime" for XM_021568071	Peroxisomal acyl-coenzyme A oxidase 1, first enzyme in the β-oxidation.
Actin	5´- GG G GCA GTA TGG CTT GTA TG-3´	5´-CT G GCA CCC TAA TCA CCT CT-3´	Albertsson et al., 2009	Housekeeping gene. Multifunctional protein that for example form filaments making up parts of the cytoskeleton in eukaryotic cells.
Ubiquitin	5´- ACA ACA TCC AGA AAG AGT CCA- 3	5´- AGG CGA GCG TAG CAC TTG-3´	Carney Almroth et al., 2008	Housekeeping gene. Regulatory protein. Marks proteins for degradation or other activities.
Bought from:	Eurofins Genomics			

References:

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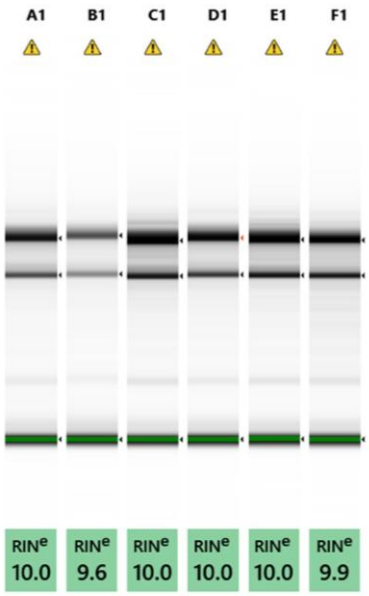
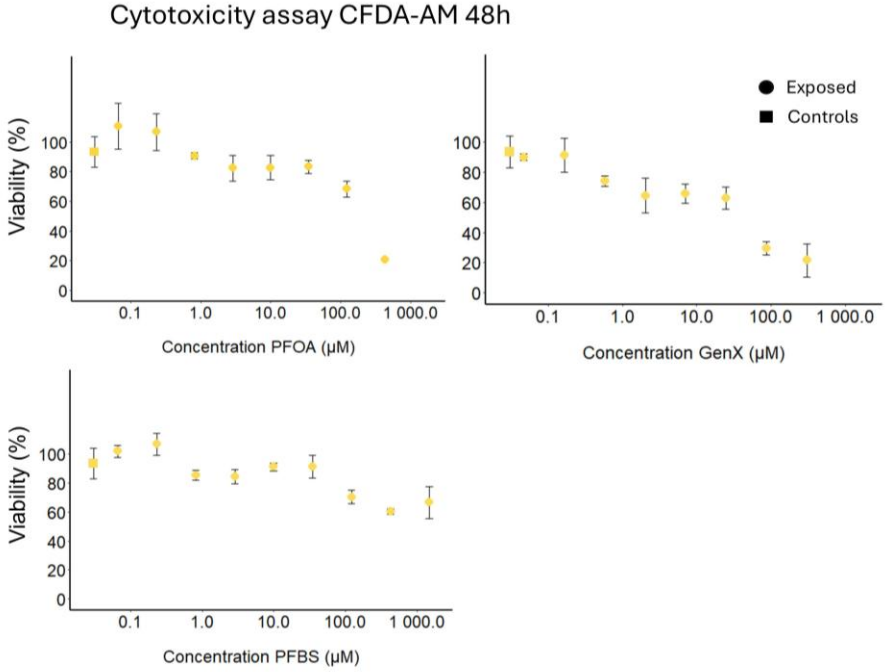
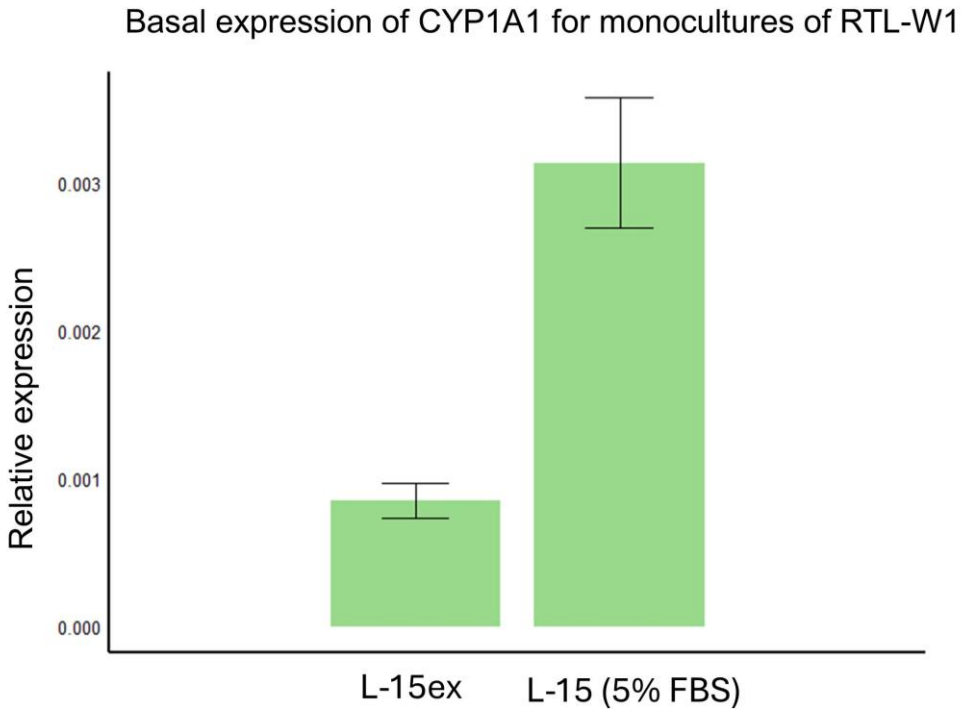
Appendix 2. Tape station confirmation of RNA quality



Well	RINe	28S/18S (Area)	Conc. [ng/µl]	Sample Description
A1	10.0	2.3	59.2	C3
B1	9.6	1.9	35.1	C4
C1	10.0	2.9	93.3	L3
D1	10.0	2.3	62.8	L4
E1	10.0	2.3	82.7	H3
F1	9.9	2.1	76.2	H4

Tape station data. Tape station gel kit run on the tapestation reader for six random RNA samples. The ribosomal RNA ratio (28S/18S) above 2.0 indicates excellent RNA quality.

Appendix 3. CFDA-AM assay data plotted with error bars.



Appendix 4. Comparison of basal CYP1A1 expression for monocultures of RTL-W1 in FBS-supplemented vs minimal medium